



Faculty of Engineering and the Built Environment: DEECE
(Department of Electrical, Electronic and Computer
Engineering)

OPERATING SYSTEMS

Supervirsor: MR. MOYO

TOPIC: WINDOWS SERVER 2019

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ALWANDE MTEMBU

EBDILL NIMBONA

DUE DATE: 23/08/2024

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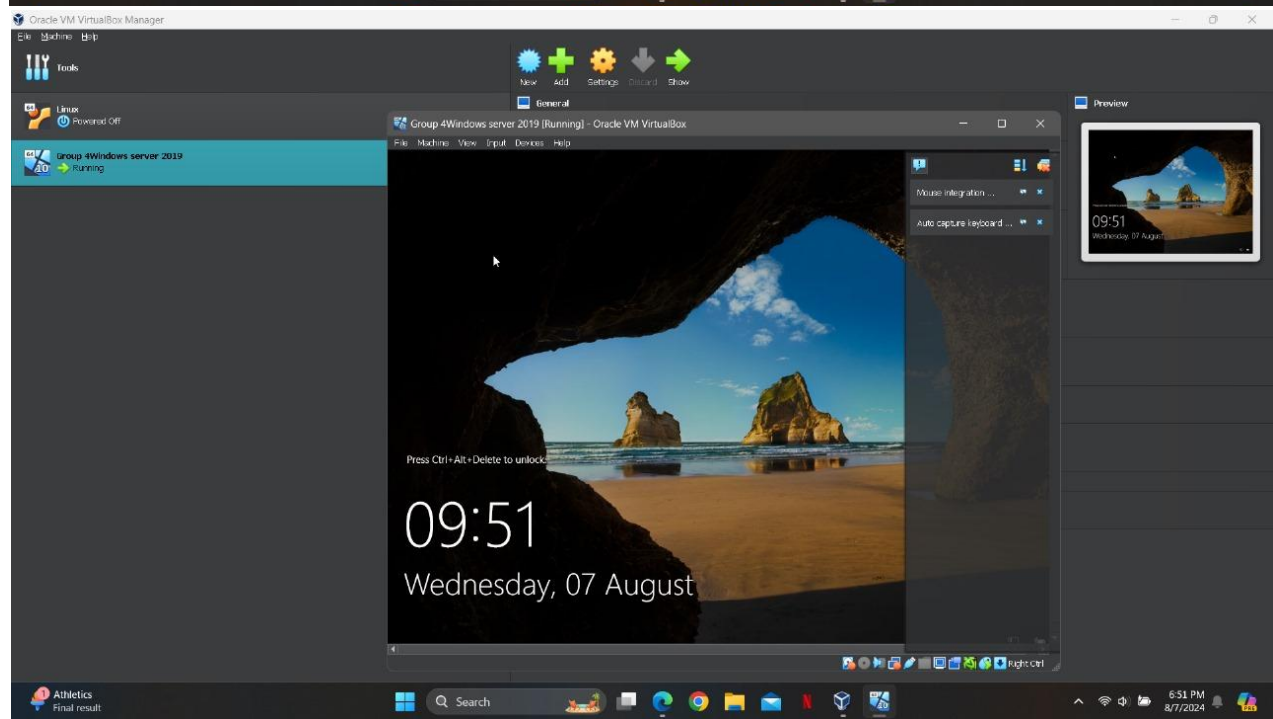
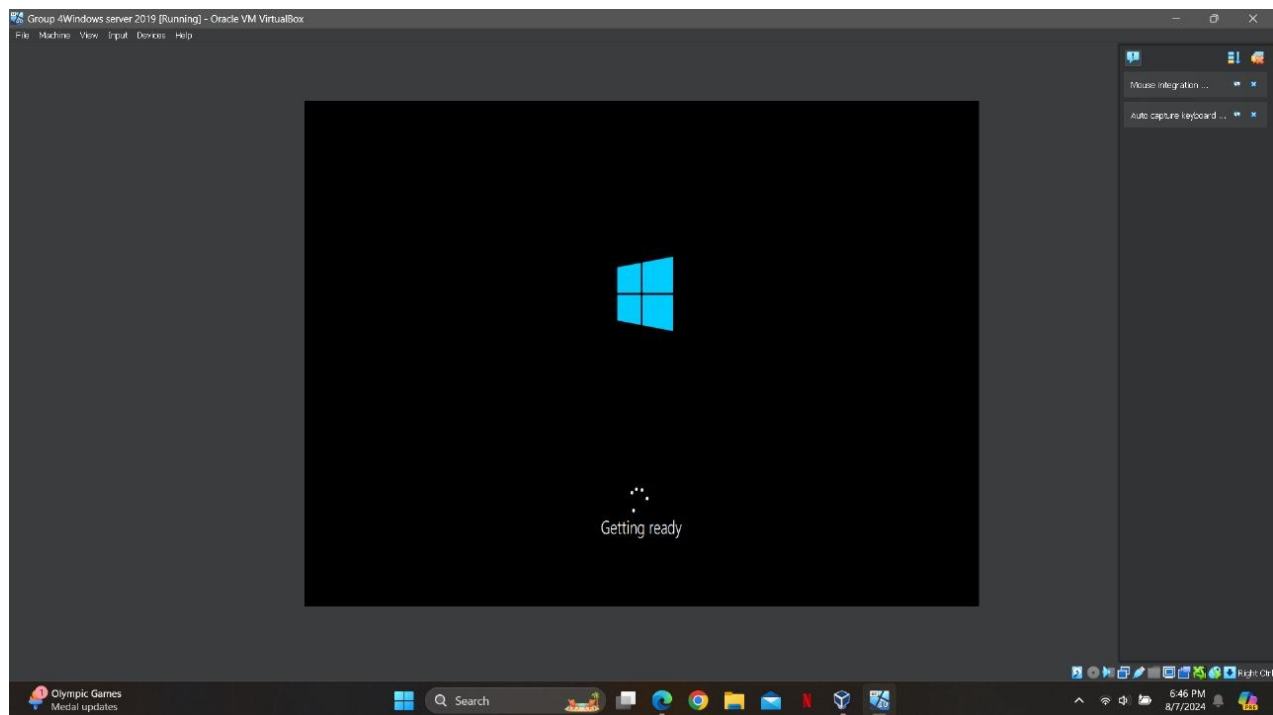
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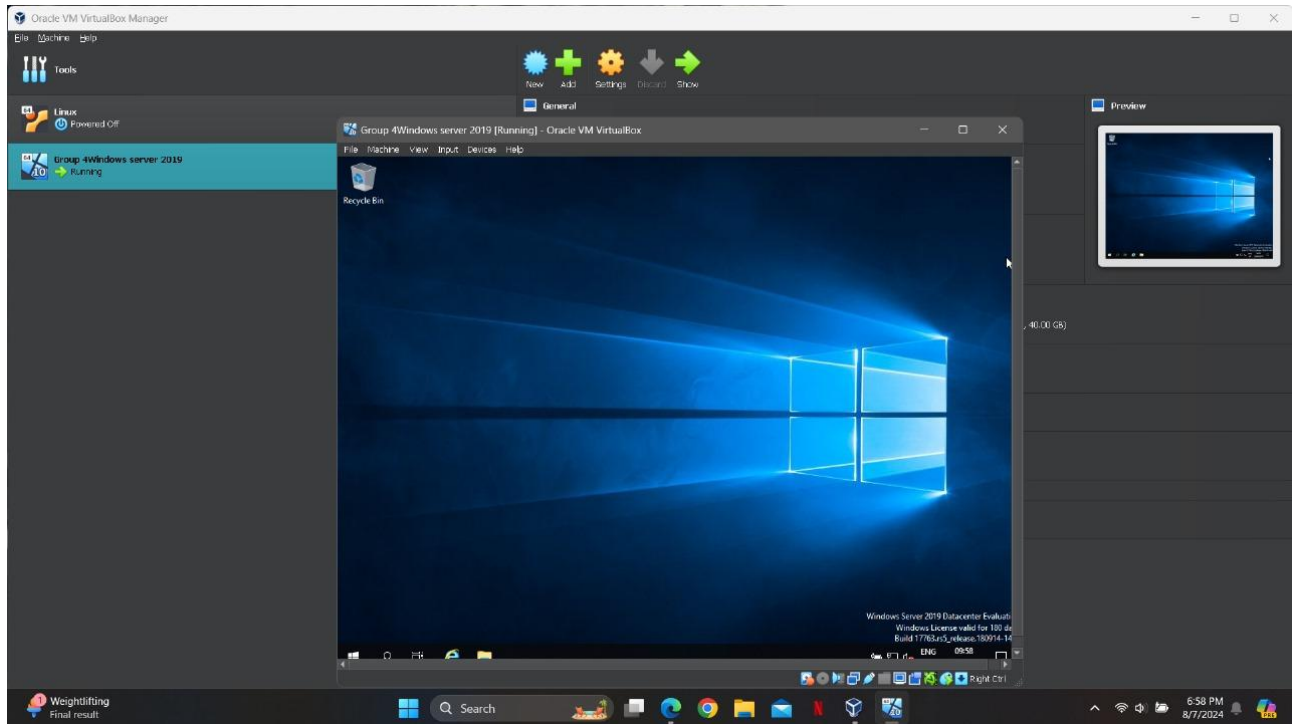
1.Introduction

A task was given to design and implement a Windows Server 2019 environment that covers key services such as Active Directory, DNS, DHCP, Group Policies, Remote Access, and Window Certificates Services. Research will be conducted using several resources to further understand and implement the server installations and configurations(edit).

2.Server installation

Server Installations refers to the restriction on how many separate computers you can install the software on. Each installation on a different computer is counted separately. A backup server ready to take over immediately counts as another installation, but installations for training, development, or inactive backups do not (double check).



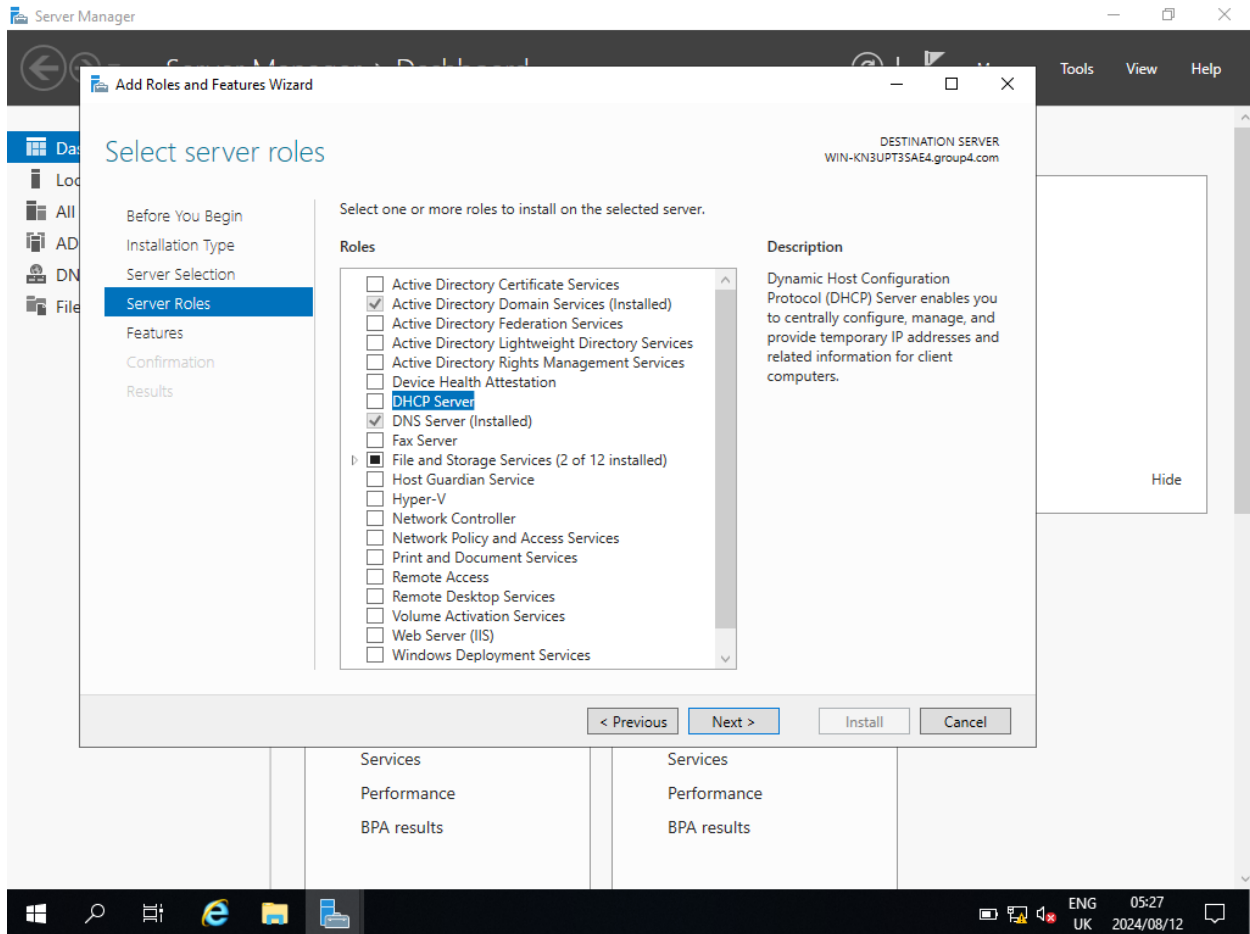


3.ADDS configuration

An Active Directory Domain Services is a server in the Active Directory that allows administrators to manage and store information about resources from a network and application data in a distributed database. The Active Directory is a directory that serves as a database which stores and centralizes several types of information about an organization and is hosted on a domain controller. The ADDS helps administrators to manage elements of a network (computers and end users) and reorders them into a hierarchy. ADDS also integrates security by authenticating logins and controlling who has access to the directory resources.

4.DHCP configuration

A Dynamic Host Configuration Protocol is a server network protocol that dynamically assigns IP addresses to network devices. DHCP automatically distributes and updates IP addresses and other configuration information on a network. The DHCP server provides information to the DHCP client through the exchange of a series of messages known as DHCP transaction.



Server Manager

Server Manager ▸ Dashboard

Manage Tools View Help

File Action View Help

DHCP

win-kn3upt3sae4.group4.com

Contents of DHCP

win-kn3upt3sae4.group4.com

Status

Actions

DHCP

More Actions

Roles: 4 | Server groups: 1 | Servers total: 1

AD DS1

Manageability

Events

Services

Performance

BPA results

DHCP1

Manageability

Events

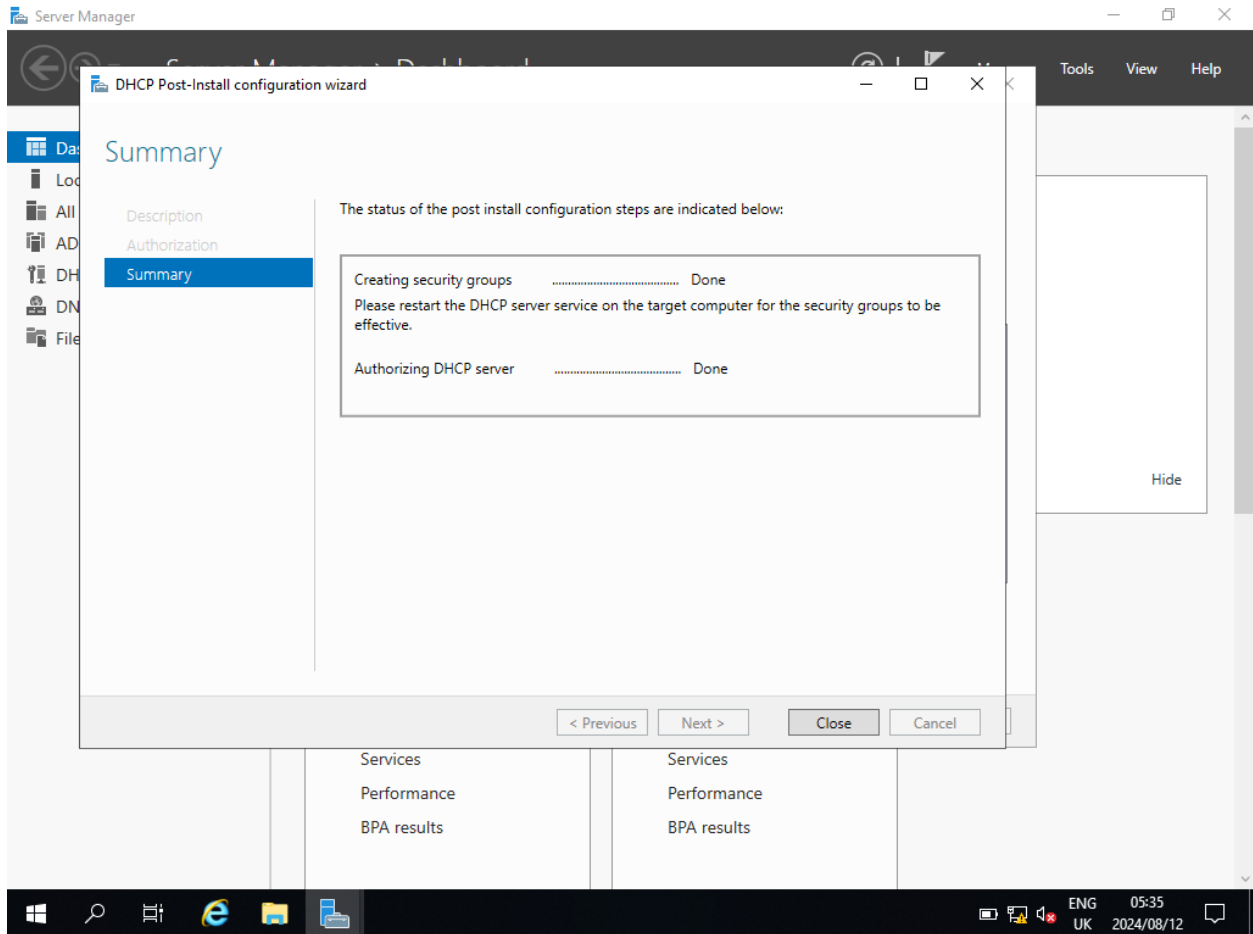
Services

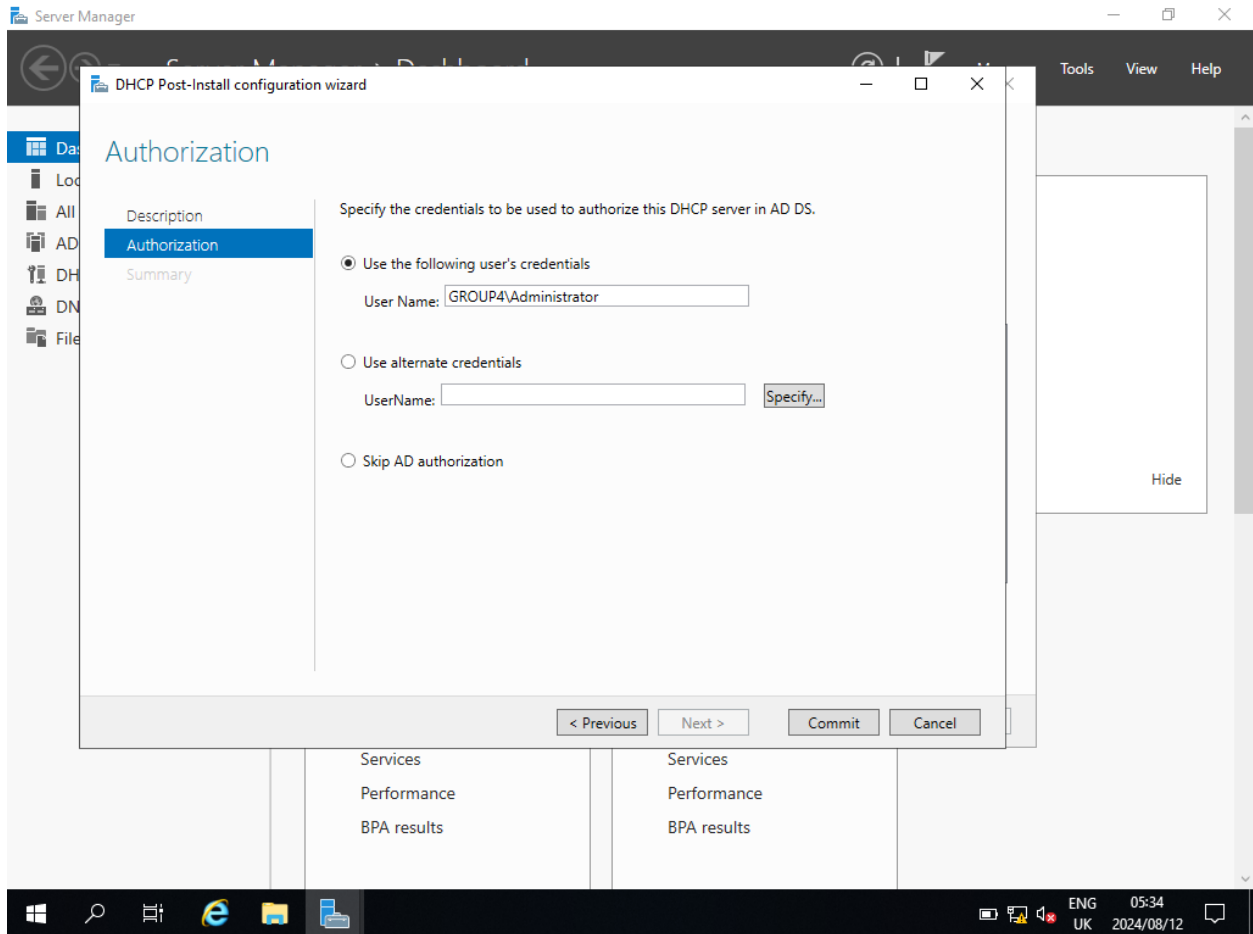
Performance

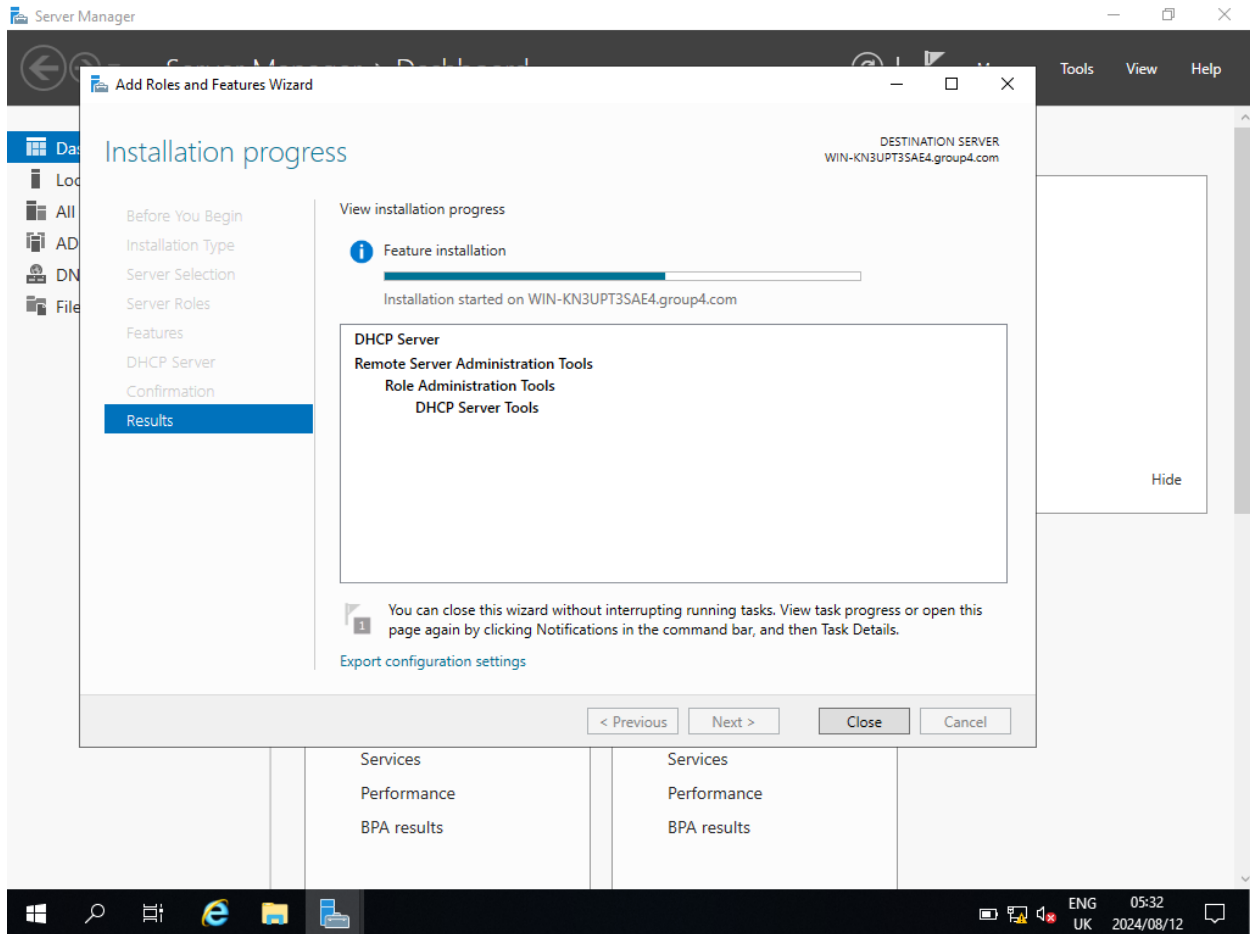
BPA results

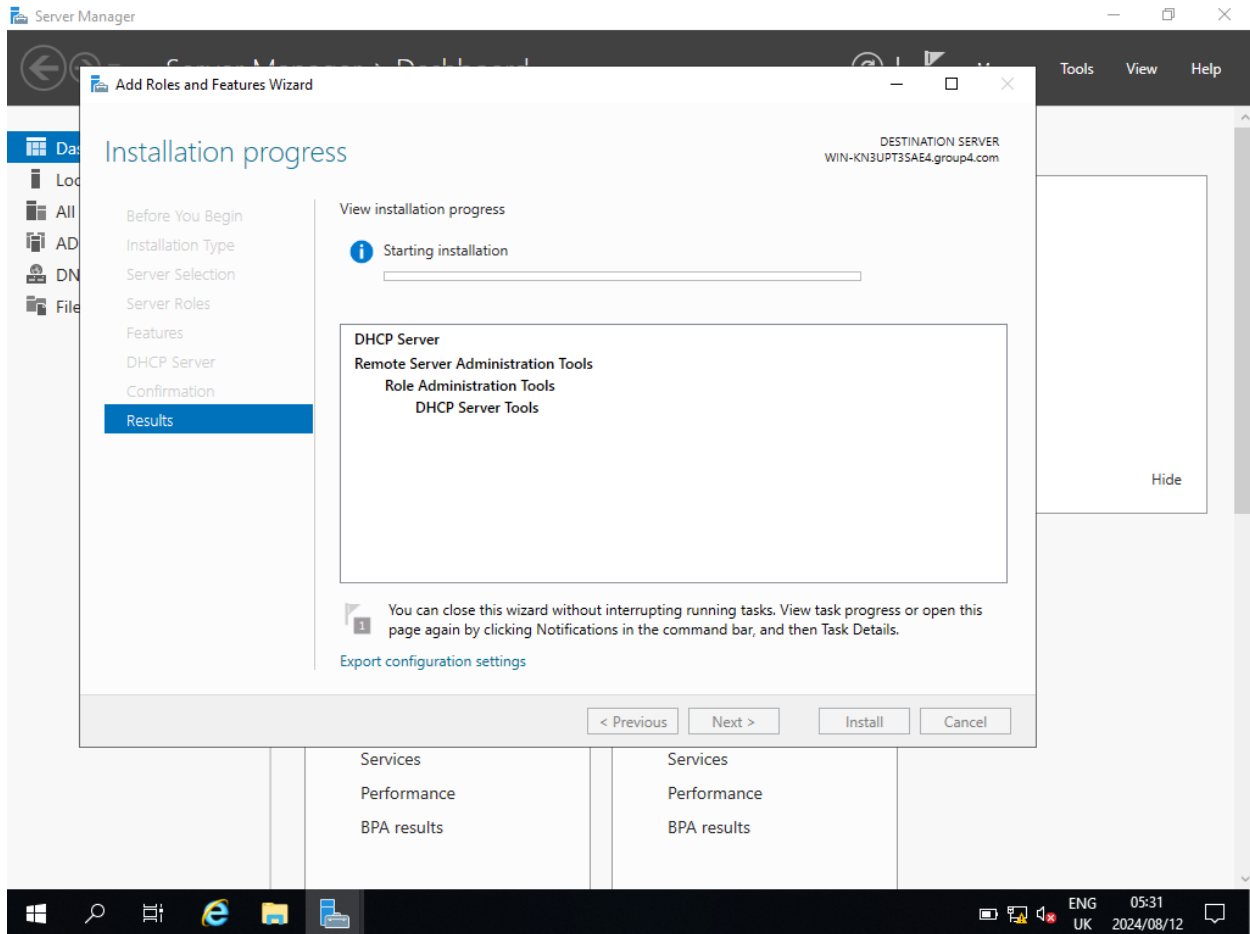
Windows Taskbar

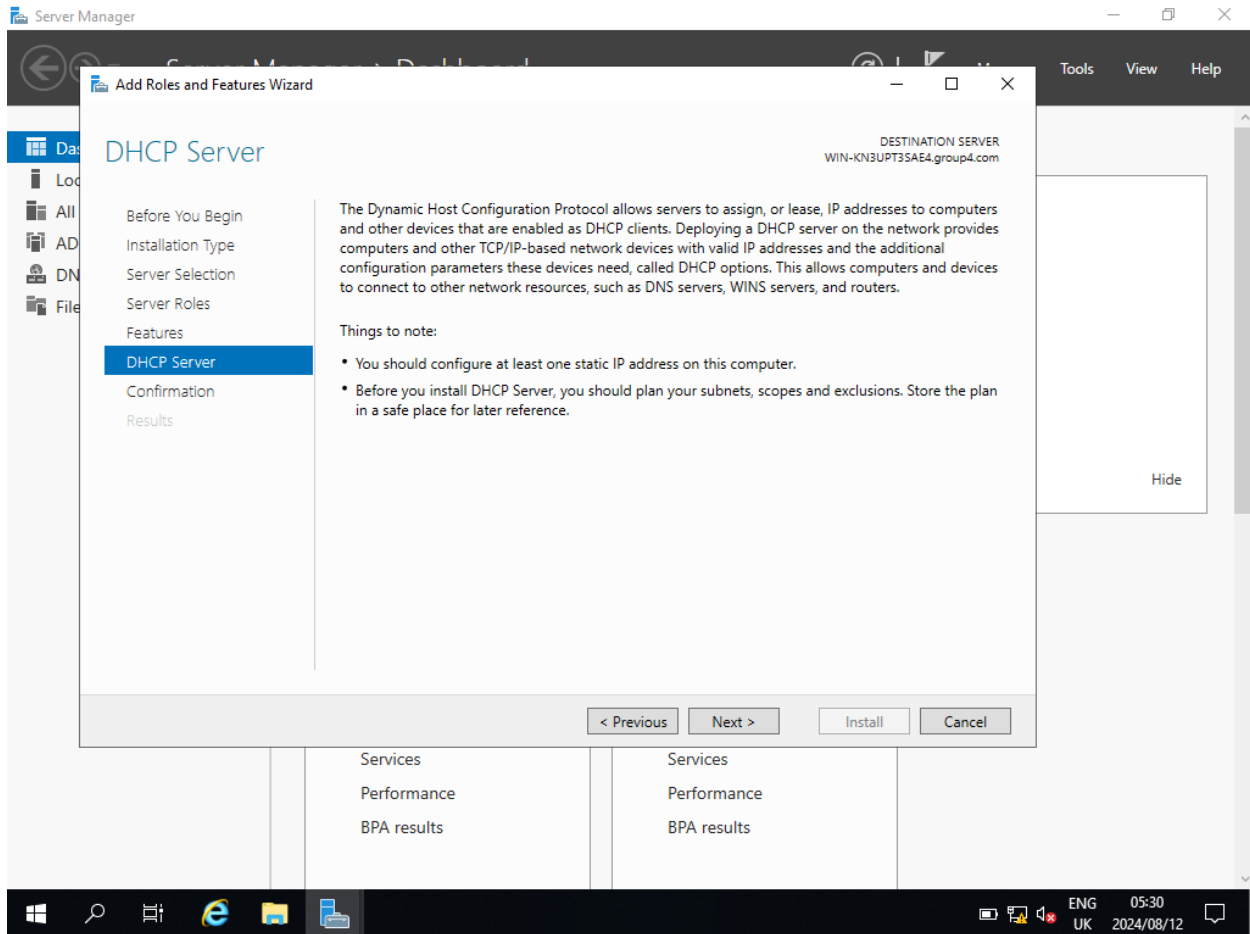
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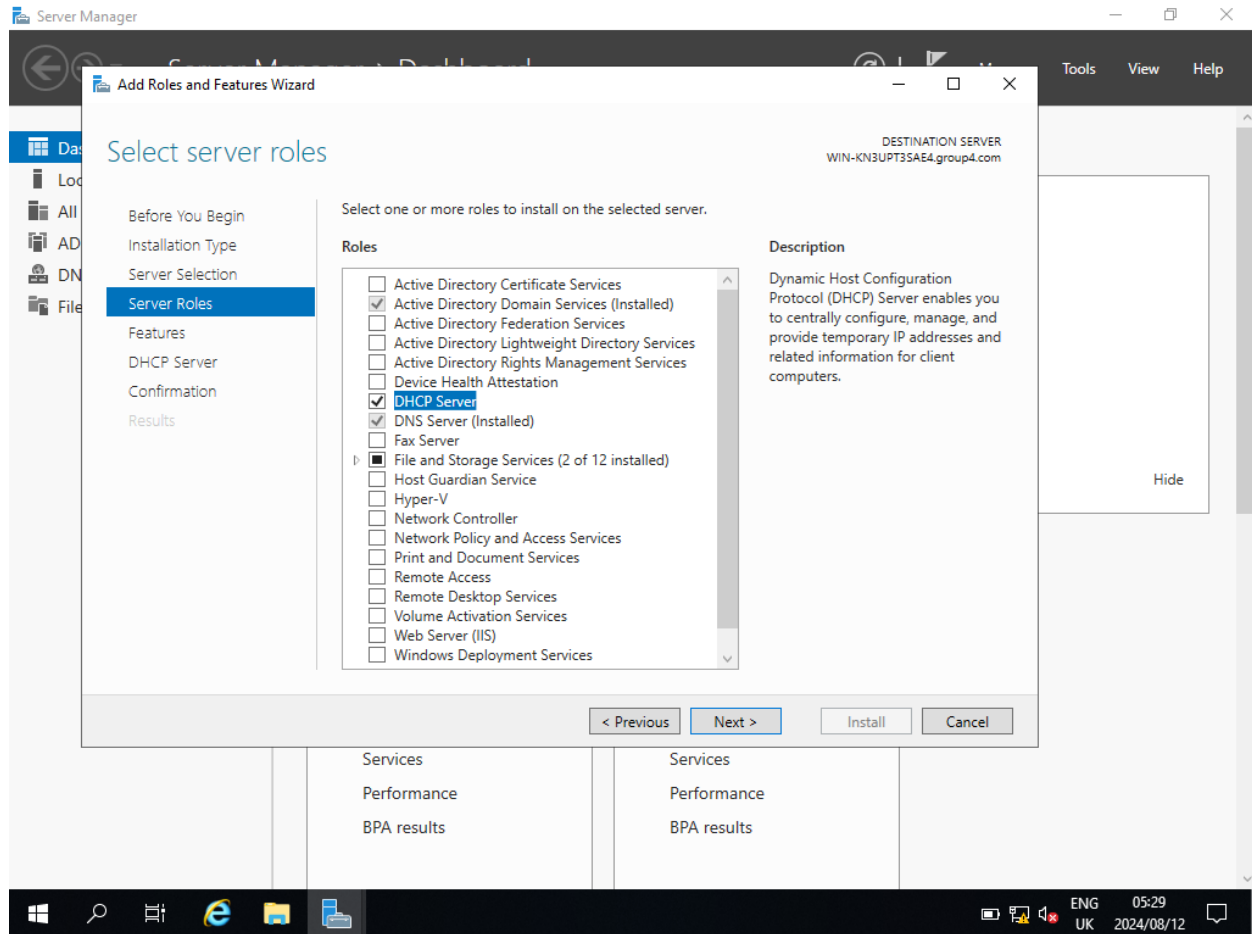






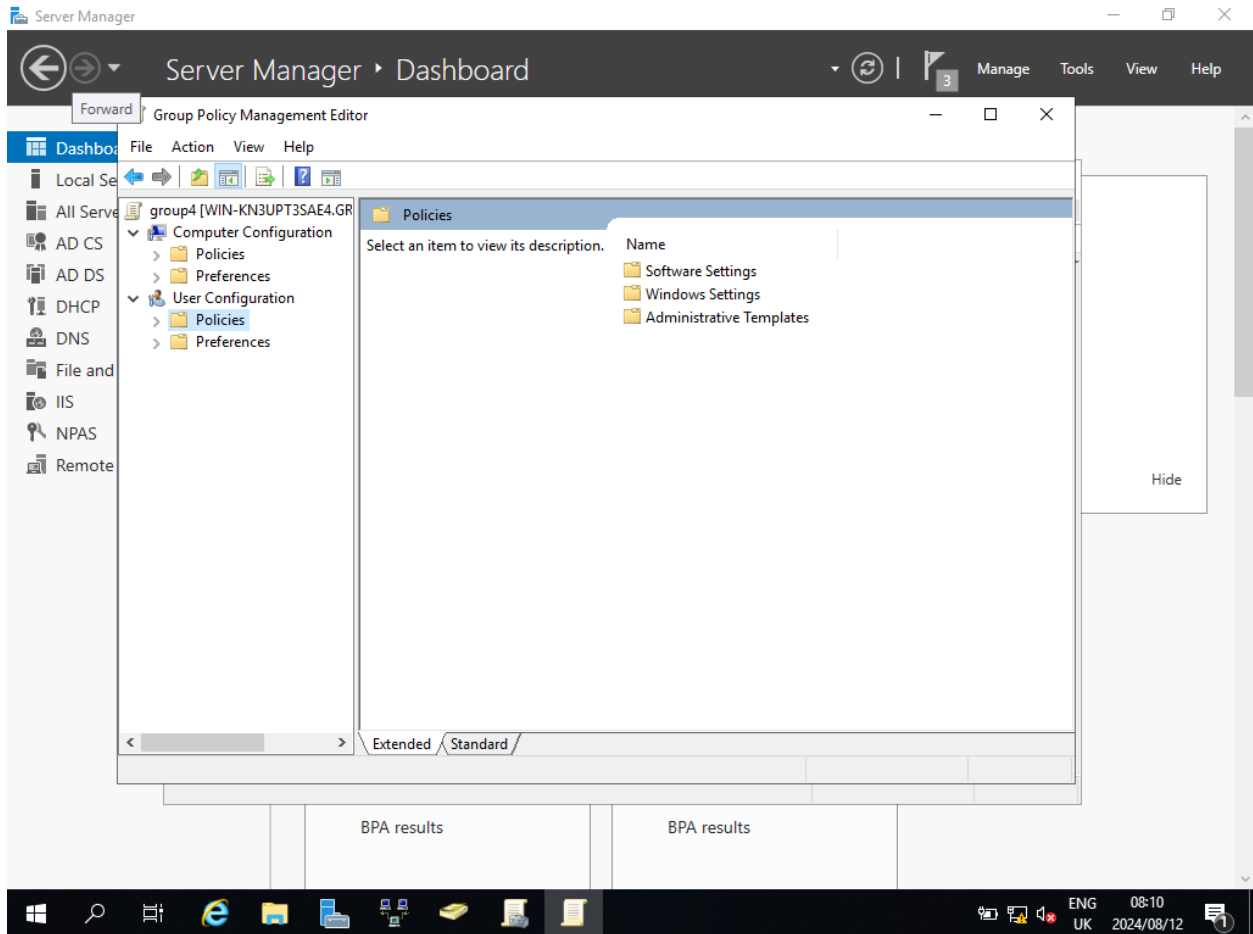


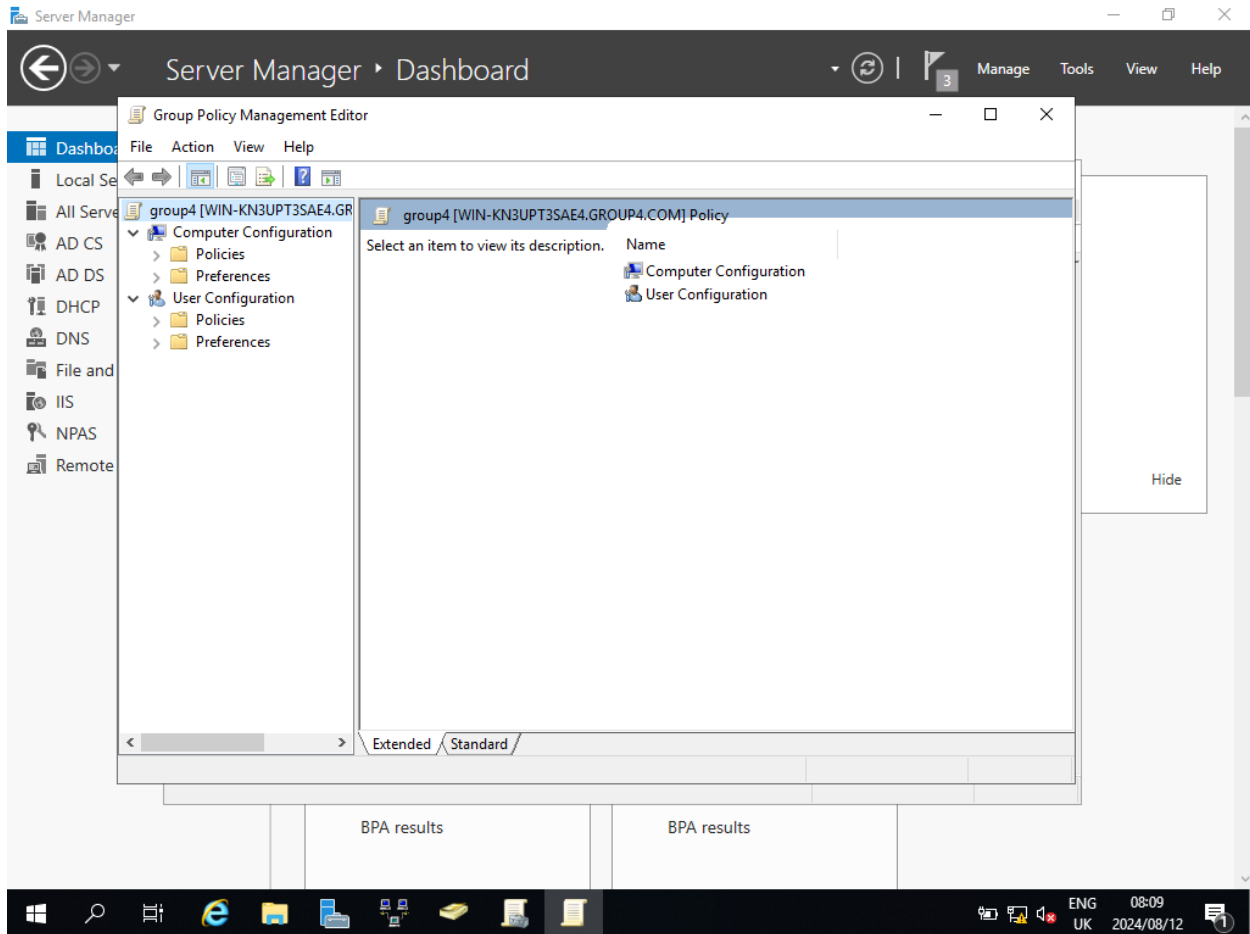




5.Group Policies configuration

Group Policies represent policy settings locally in the file system or in Active Directory Domain Services. Group Policies are contained in a Group Policy Object (GPO) when used with Active Directory. GPOs (Group Policy Object) are a virtual collection of policy settings and security permissions that can be applied to users and computers in Active Directory. Group Policy settings are divided into policy settings that affect a computer and policy settings that affect a user. Computer related policies specify system behavior, application settings, computer startup and shutdown scripts. User-related policies specify system behavior, application settings, user login and logoff scripts. For computers, Group policies are applied when the computer starts and for users Group Policies are applied when the user signs in.





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Remote Access

Group Policy Management

FileActionViewWindowHelp

Group Policy Management

Forest: group4.com

Domains

group4.com

Default Domain

CPUT(Bellville)

group4

Domain Control

group4

Group Policy Ob

WMI Filters

Starter GPOs

Sites

Group Policy Modeling

Group Policy Results

CPUT(Bellville)

Linked Group Policy ObjectsGroup Policy InheritanceDelegation

Link Order	GPO	Enforced	Link Enabled	GPO Status
1	group4	No	Yes	Enabled

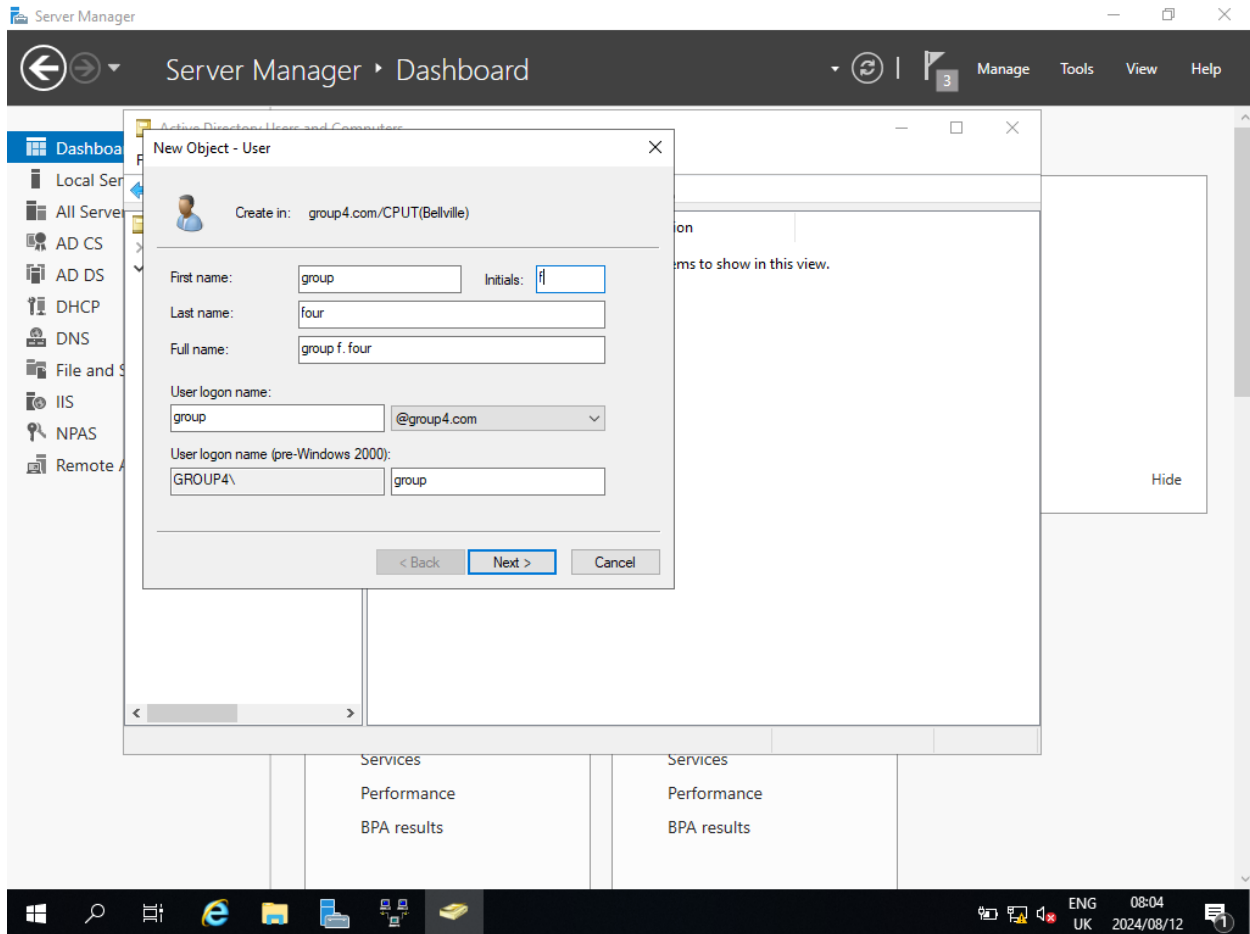
BPA results

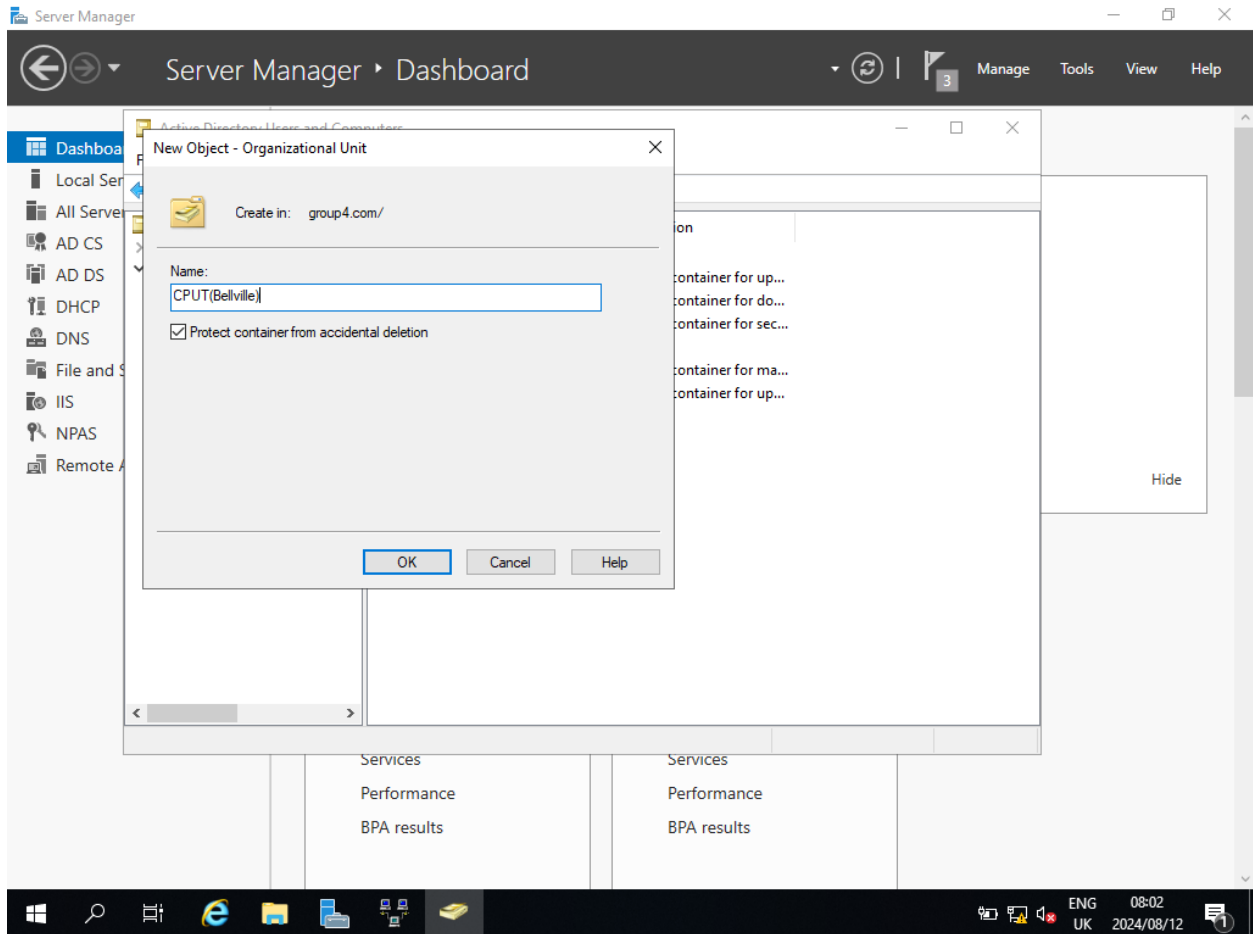
BPA results

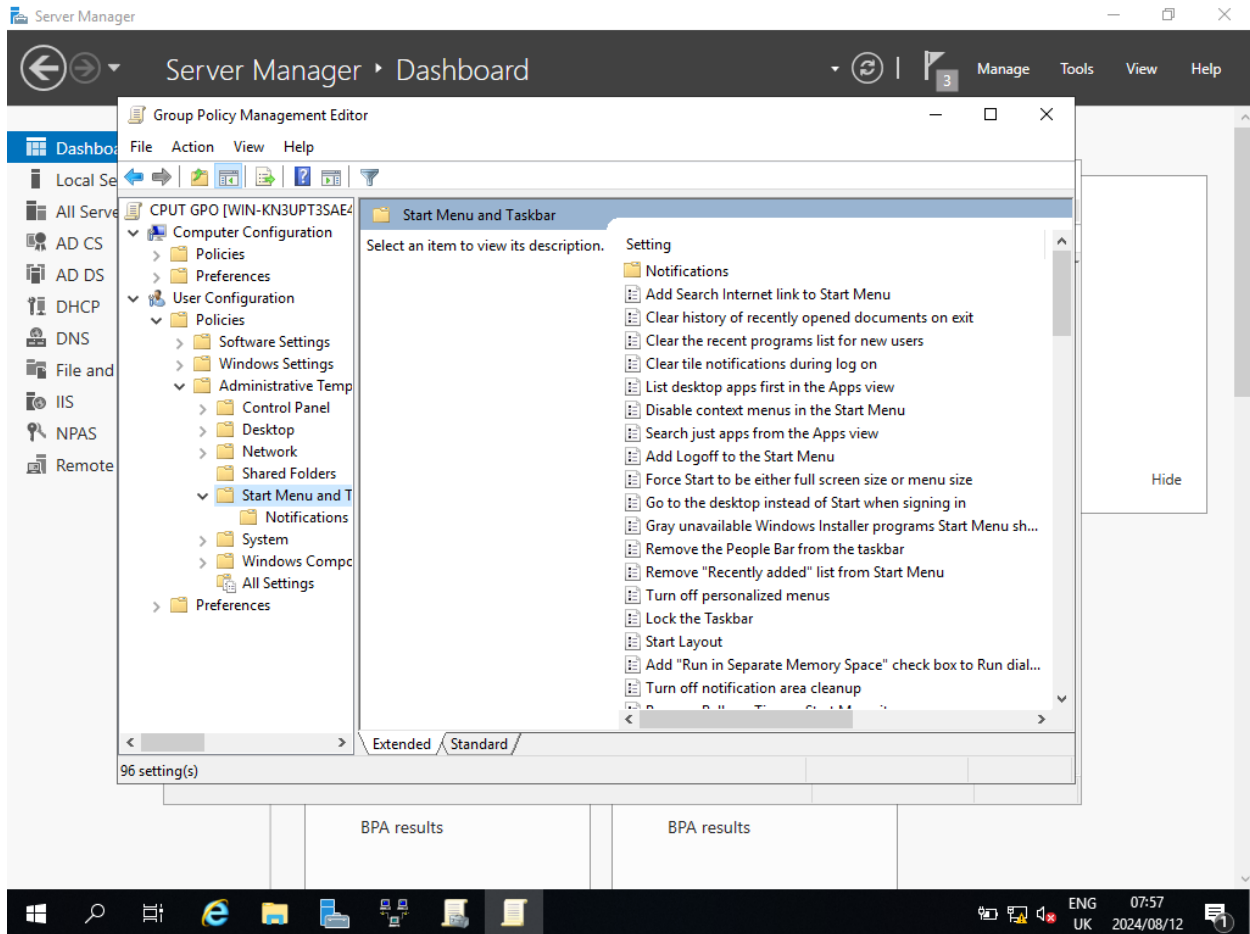
Hide

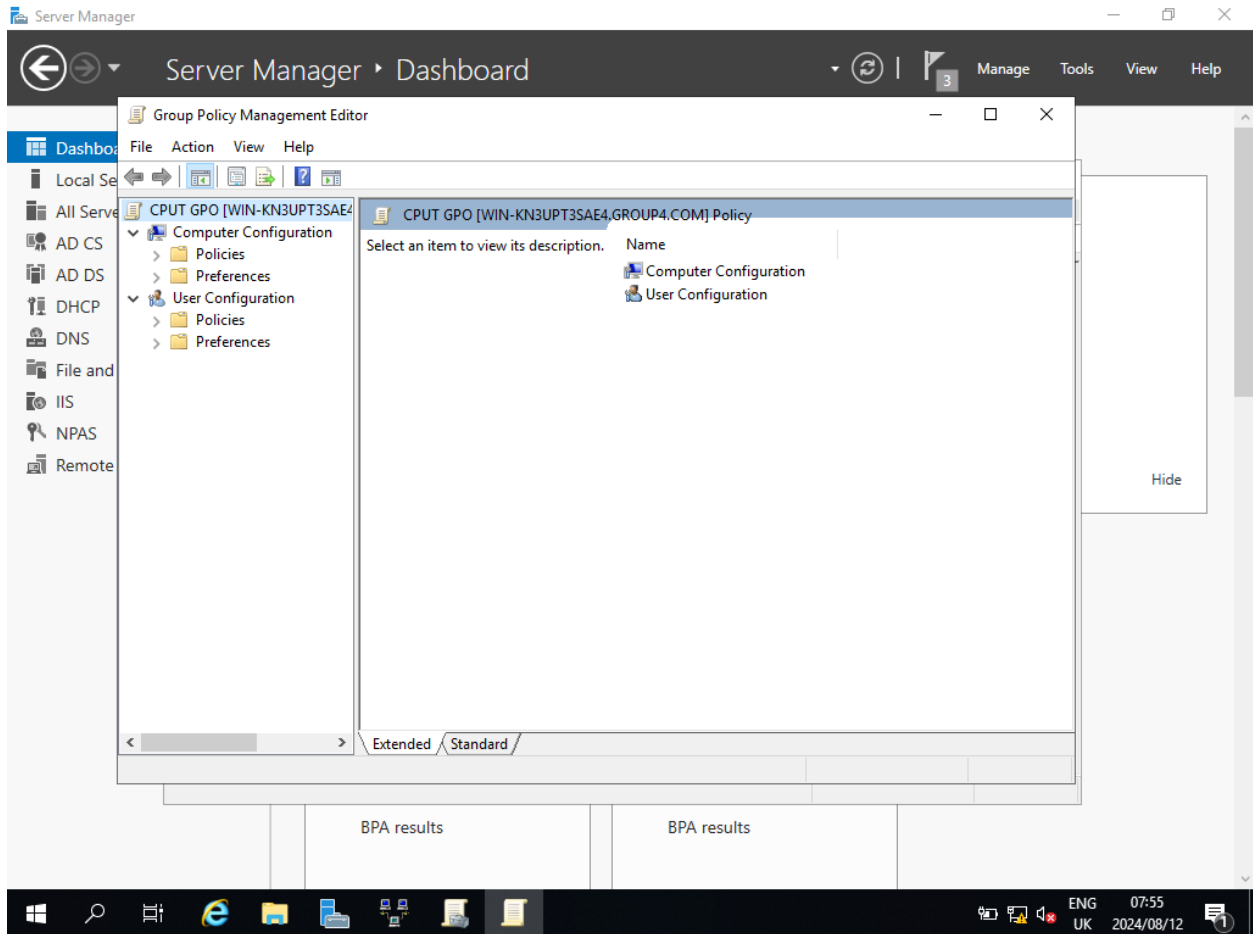
Windows Taskbar

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Group Policy Management

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Group Policy Management

Forest: group4.com

Domains

group4.com

Default Domain Controller...

group4

Group Policy Objects

Default Domain Controller...

WMI Filters

Starter GPOs

Sites

Group Policy Modeling

Group Policy Results

Group Policy Objects in group4.com

ContentsDelegation

Name	GPO Status	WMI Filter	Modified	Owner
Default Domain Controller...	Enabled	None	2024/08/12 07:...	Domain
Default Domain Controller...	Enabled	None	2024/08/12 04:...	Domain

New GPO

Name: CPUT GPO

Source Starter GPO: (none)

OKCancel

BPA resultsBPA results

Windows Taskbar

ENG UK07:542024/08/12

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group4.com

Default Domain

Domain Control

group4

Group Policy Objects

Default Domain

Default Domain

WMI Filters

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Group Policy Modeling

Group Policy Results

Group Policy Objects in group4.com

ContentsDelegation

Name	GPO Status	WMI Filter	Modified	Owner
Default Domain Controller...	Enabled	None	2024/08/12 07:...	Domain
Default Domain Policy	Enabled	None	2024/08/12 04:...	Domain

BPA results

BPA results

Hide

Windows Taskbar

Search

Taskbar Icons

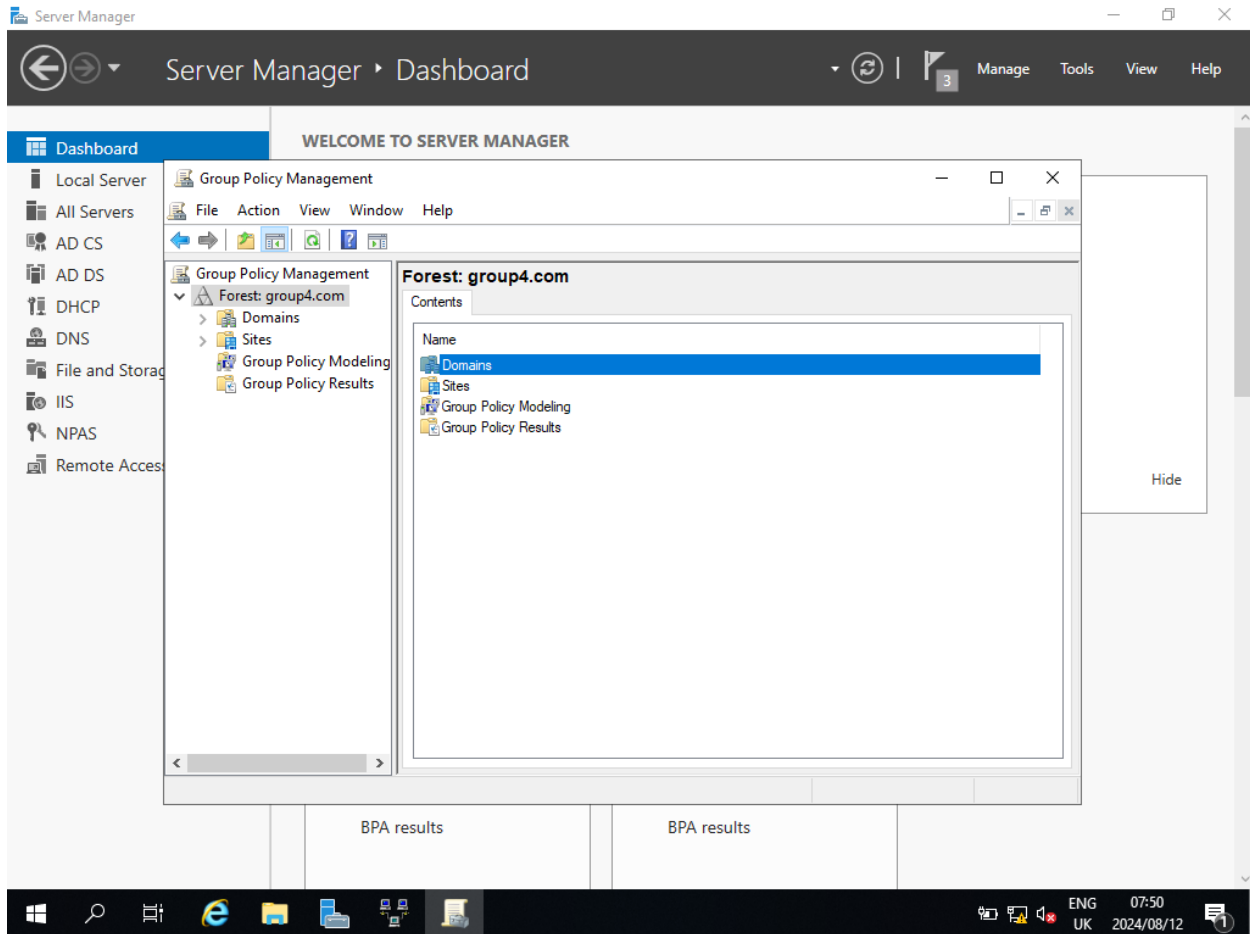
System Tray

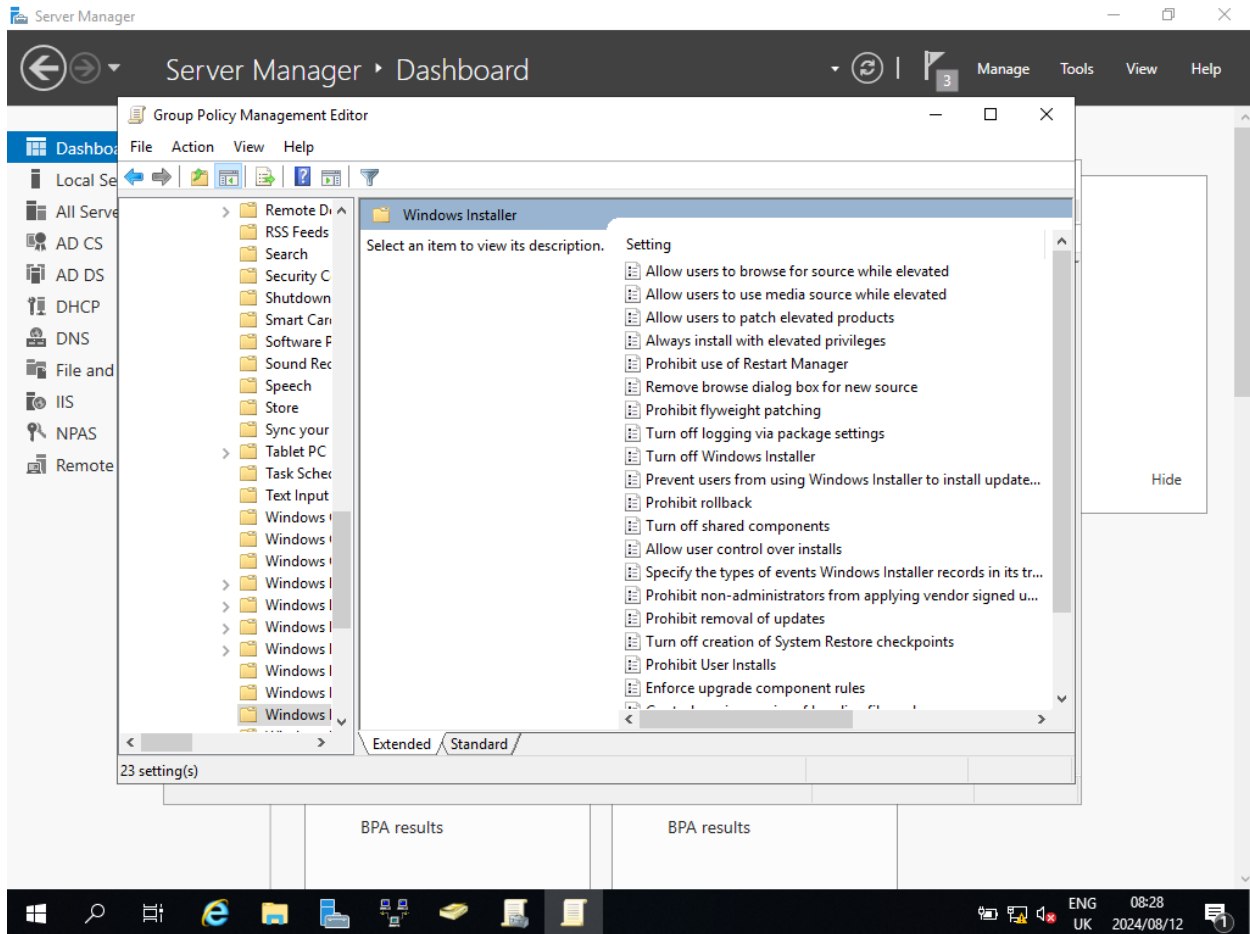
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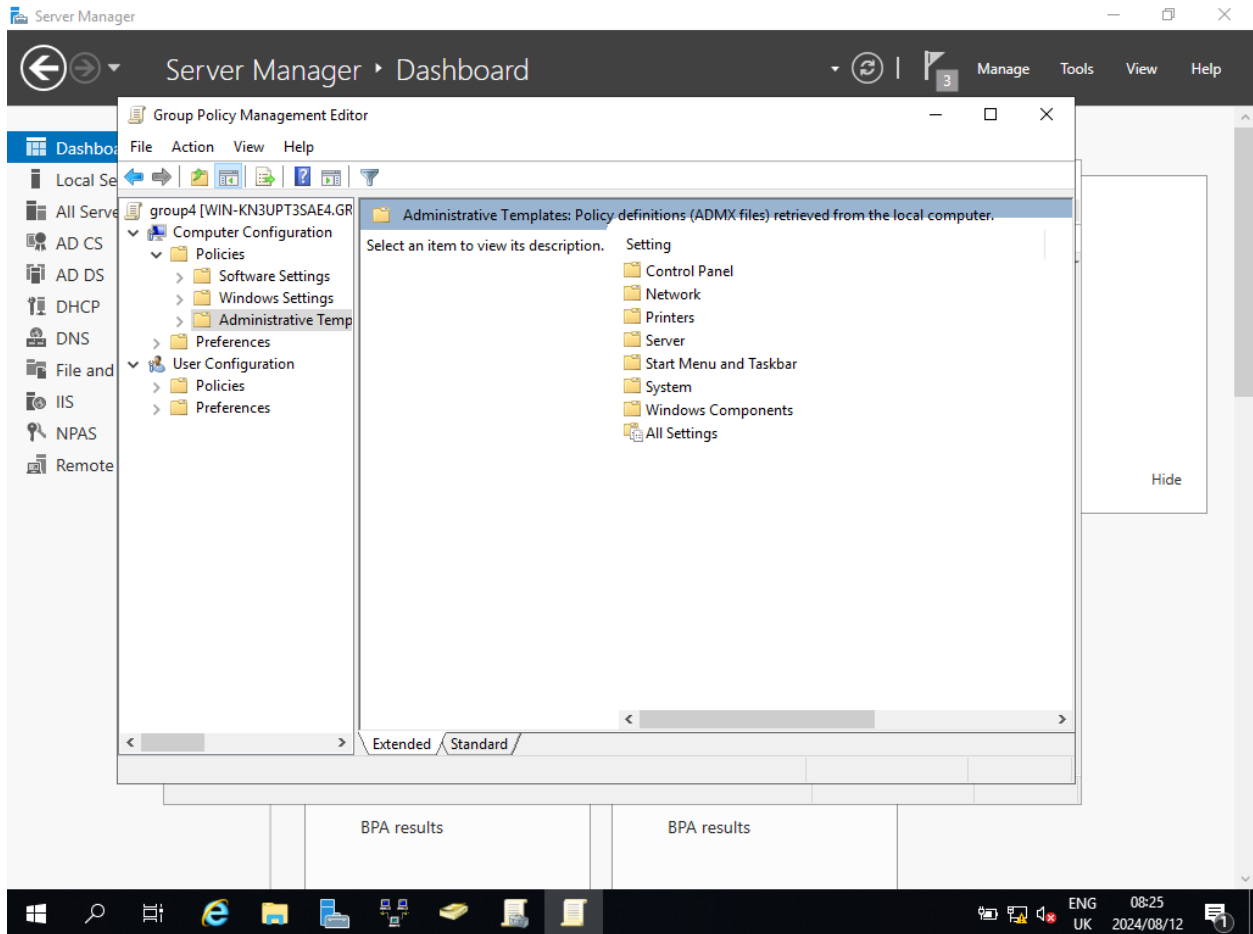
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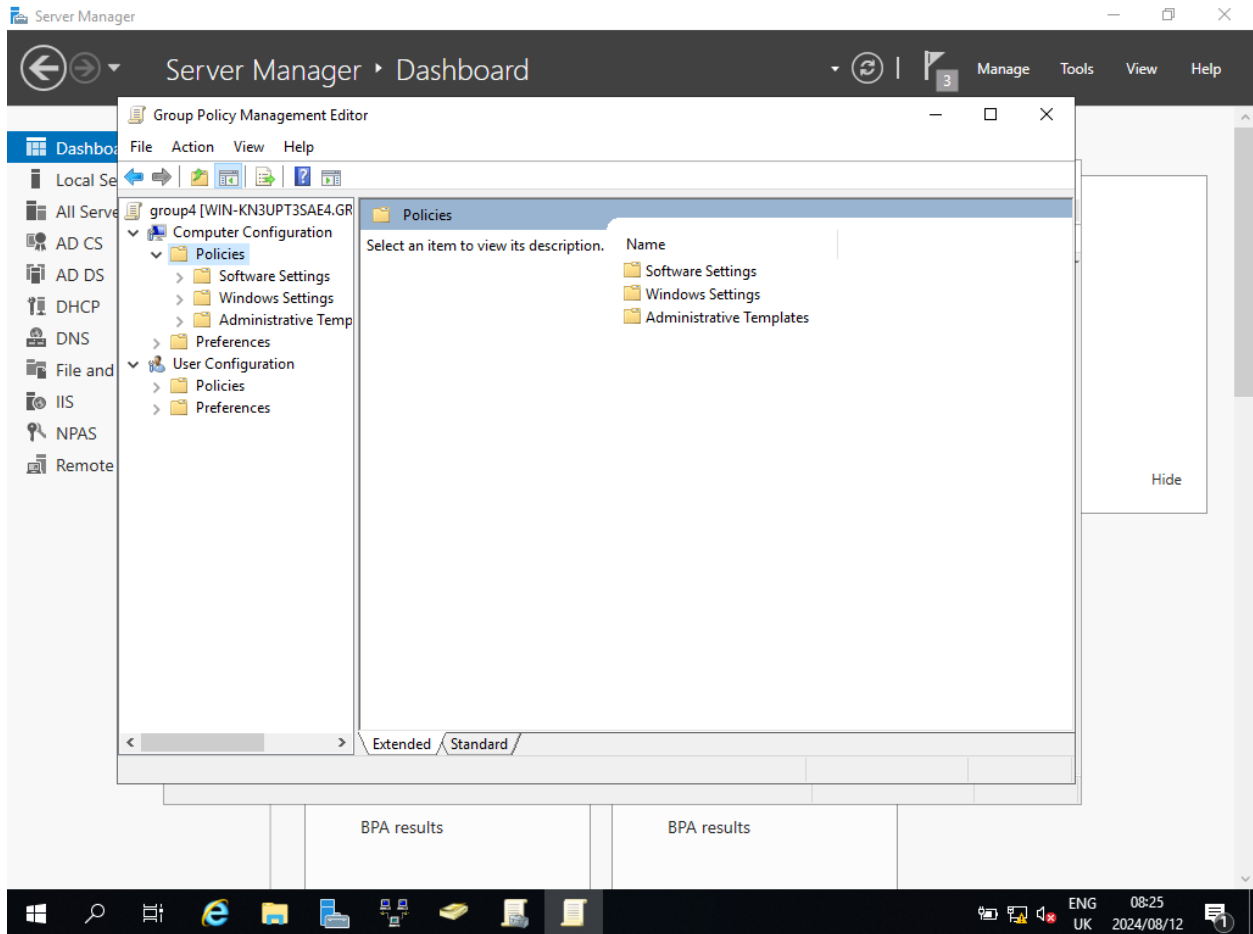
2024/08/12

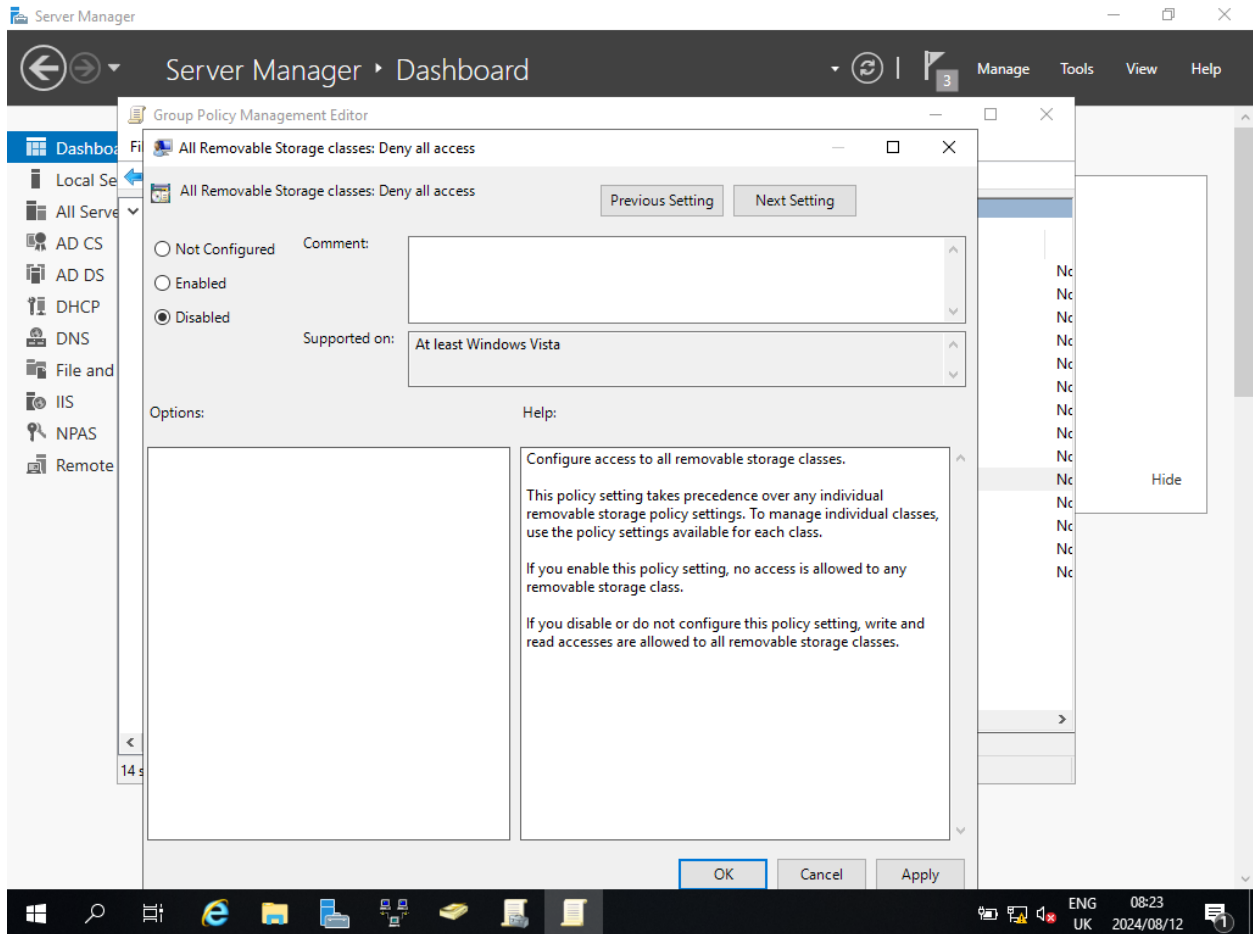
Notification

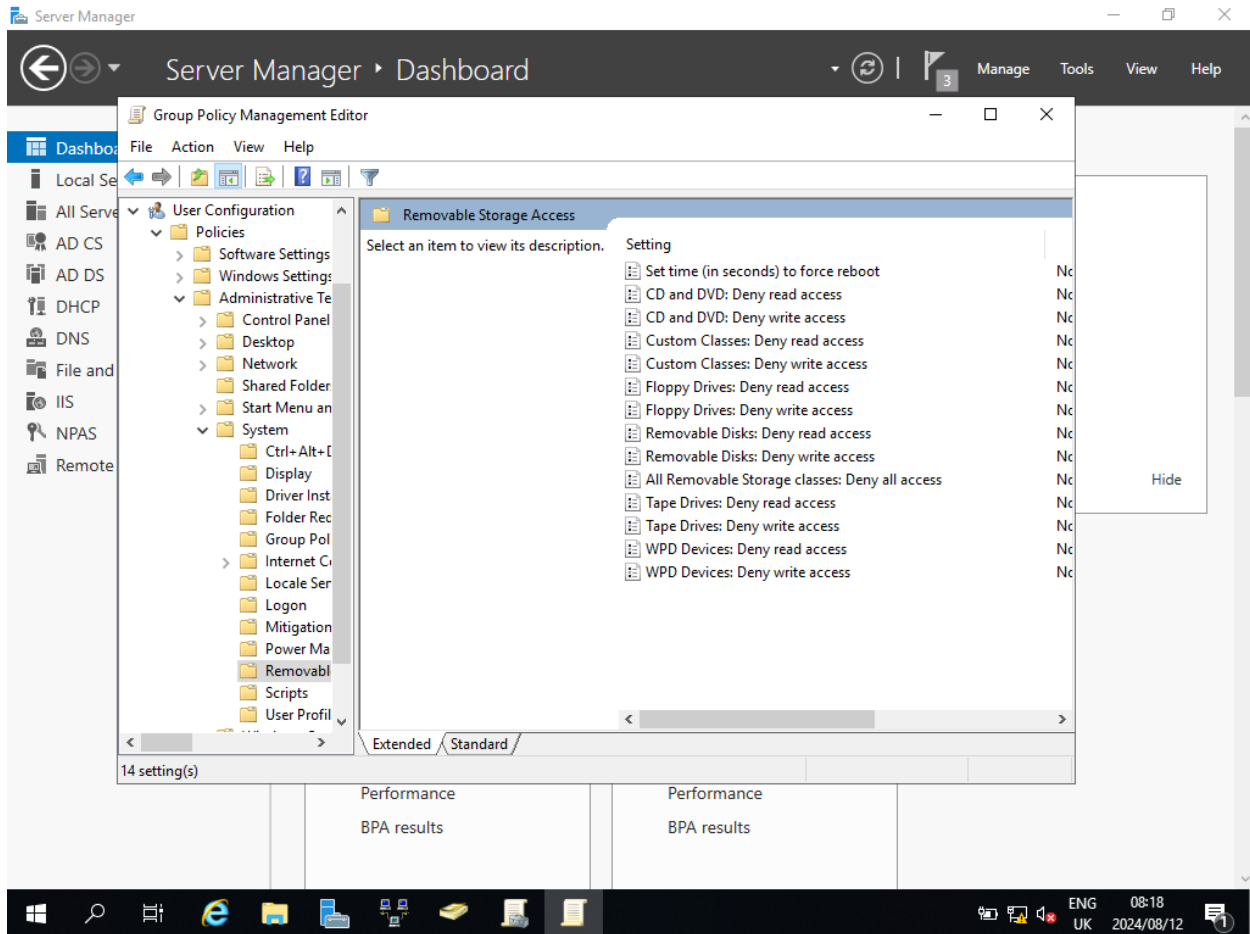


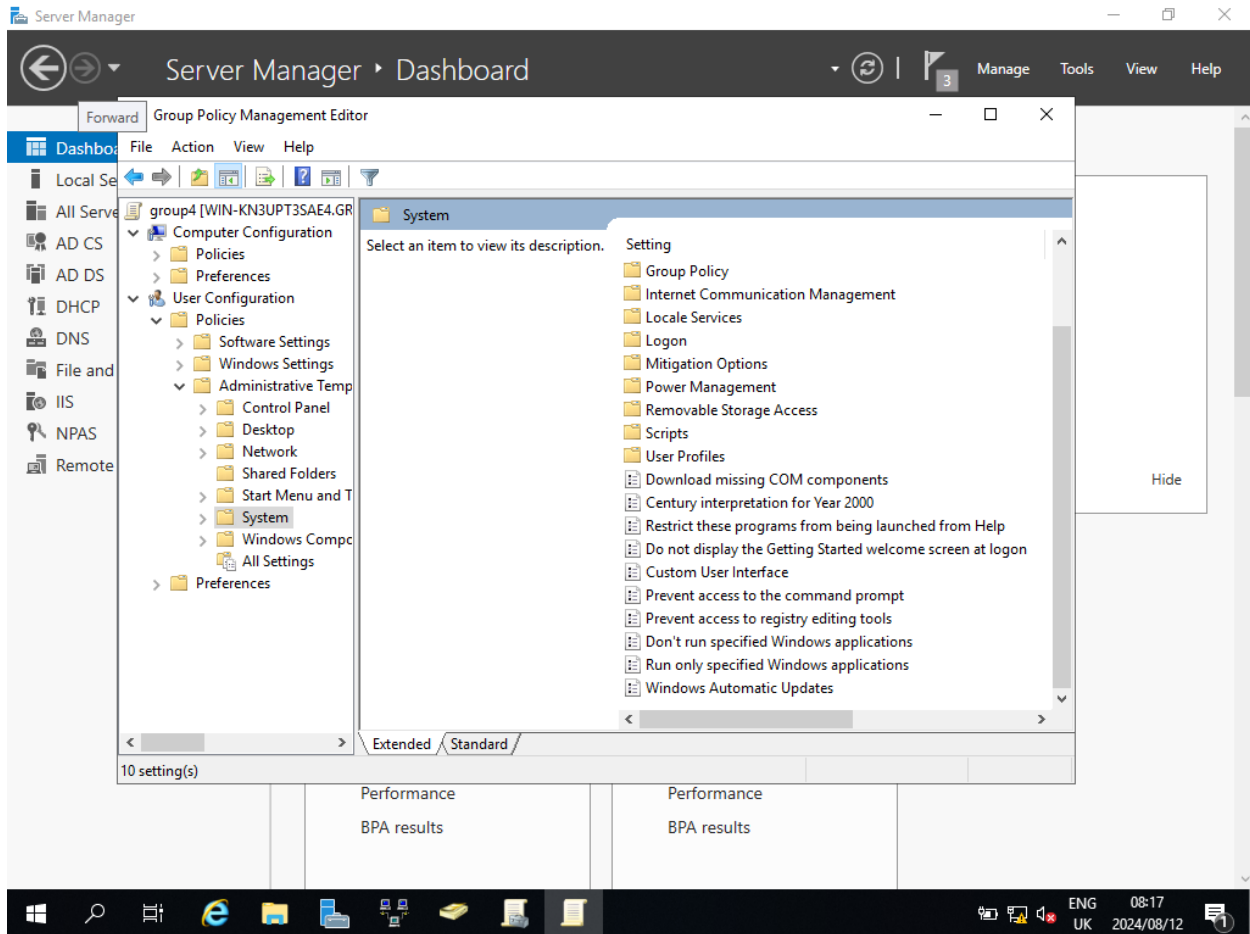


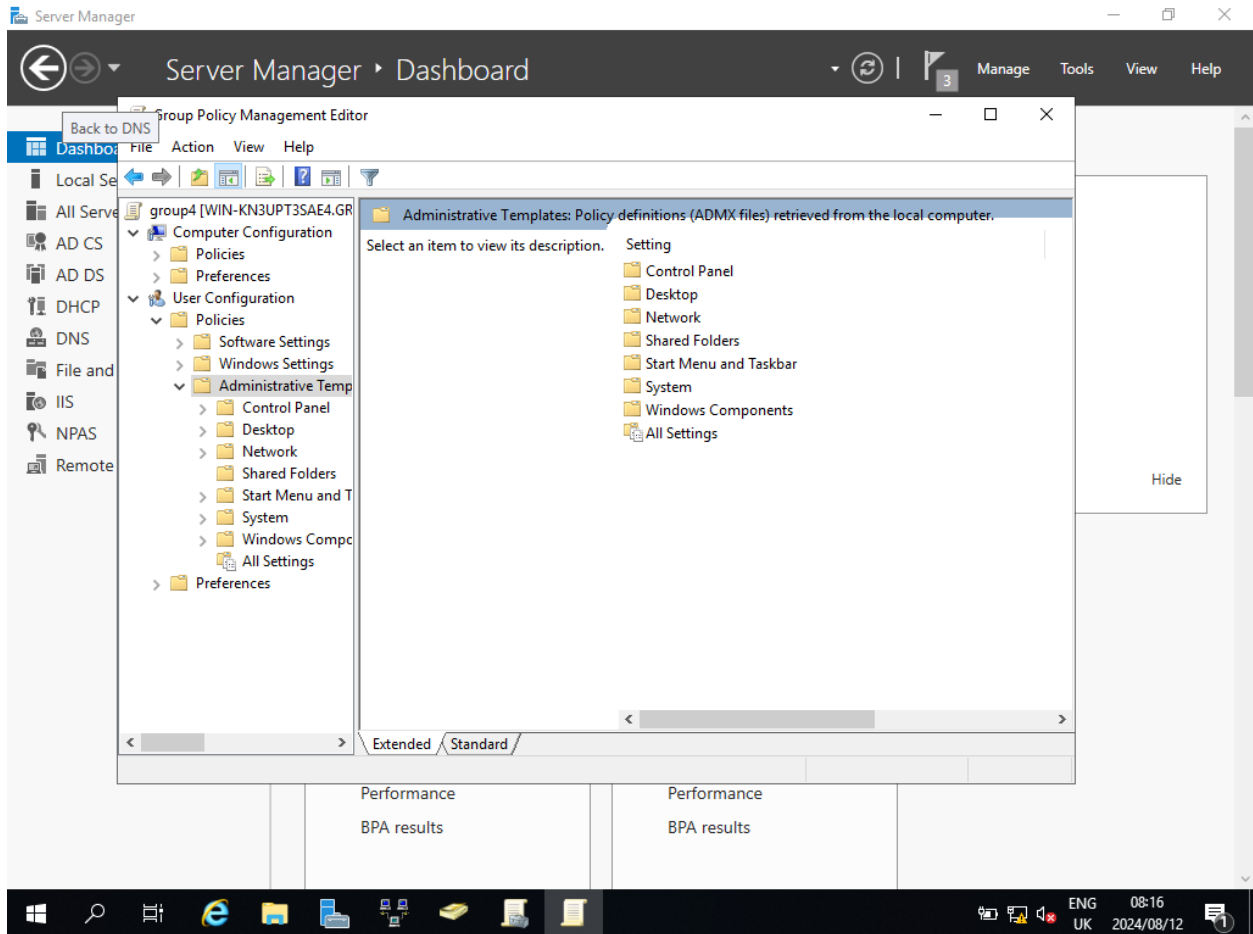












Prevent access to the command prompt

Prevent access to the command prompt

Previous Setting

Next Setting

☐ Not Configured

☐ Enabled

☒ Disabled

Comment:

Supported on:

At least Windows 2000

Options:

Help:

Disable the command prompt script processing also?

This policy setting prevents users from running the interactive command prompt, Cmd.exe. This policy setting also determines whether batch files (.cmd and .bat) can run on the computer.

If you enable this policy setting and the user tries to open a command window, the system displays a message explaining that a setting prevents the action.

If you disable this policy setting or do not configure it, users can run Cmd.exe and batch files normally.

Note: Do not prevent the computer from running batch files if the computer uses logon, logoff, startup, or shutdown batch file scripts, or for users that use Remote Desktop Services.

OK

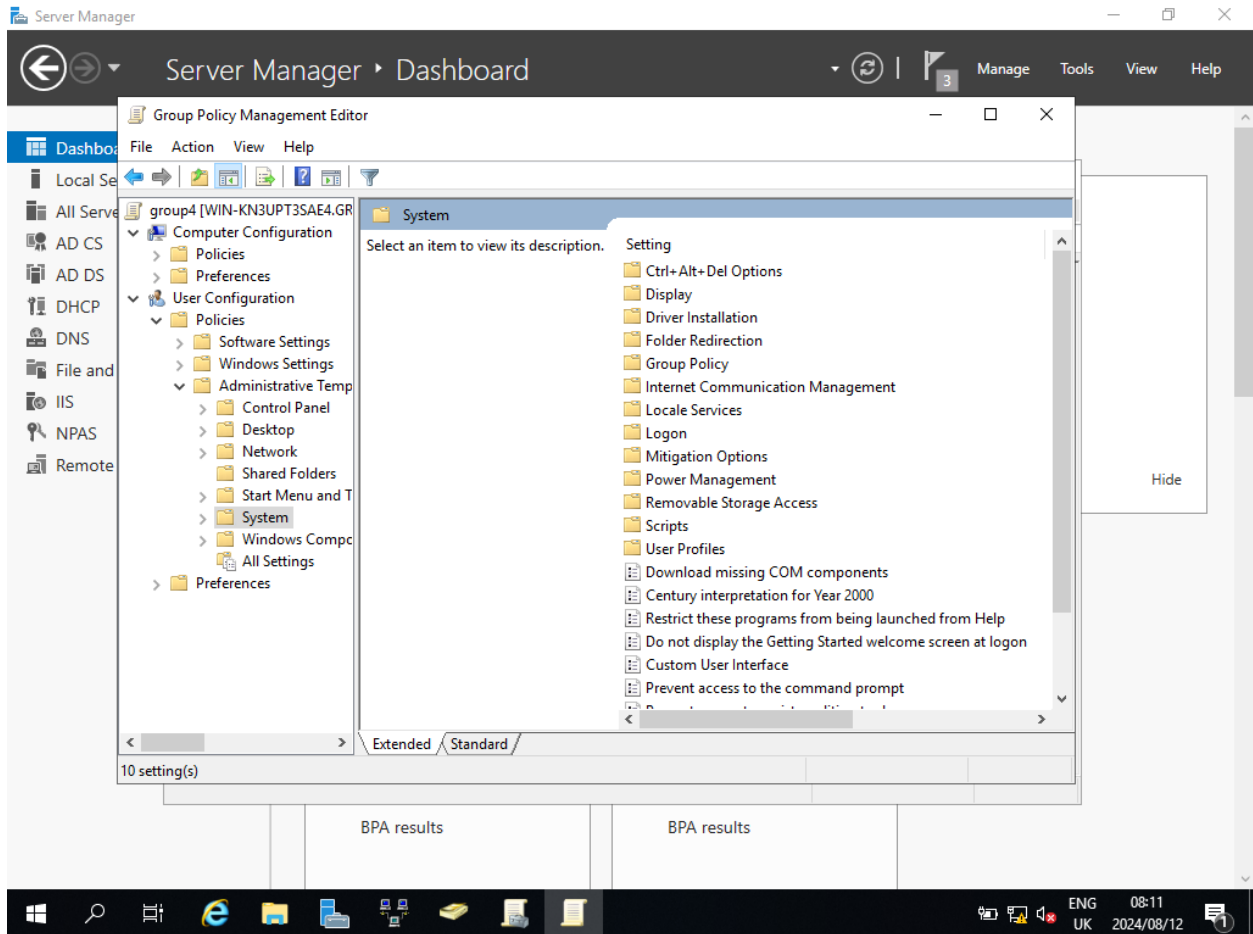
Cancel

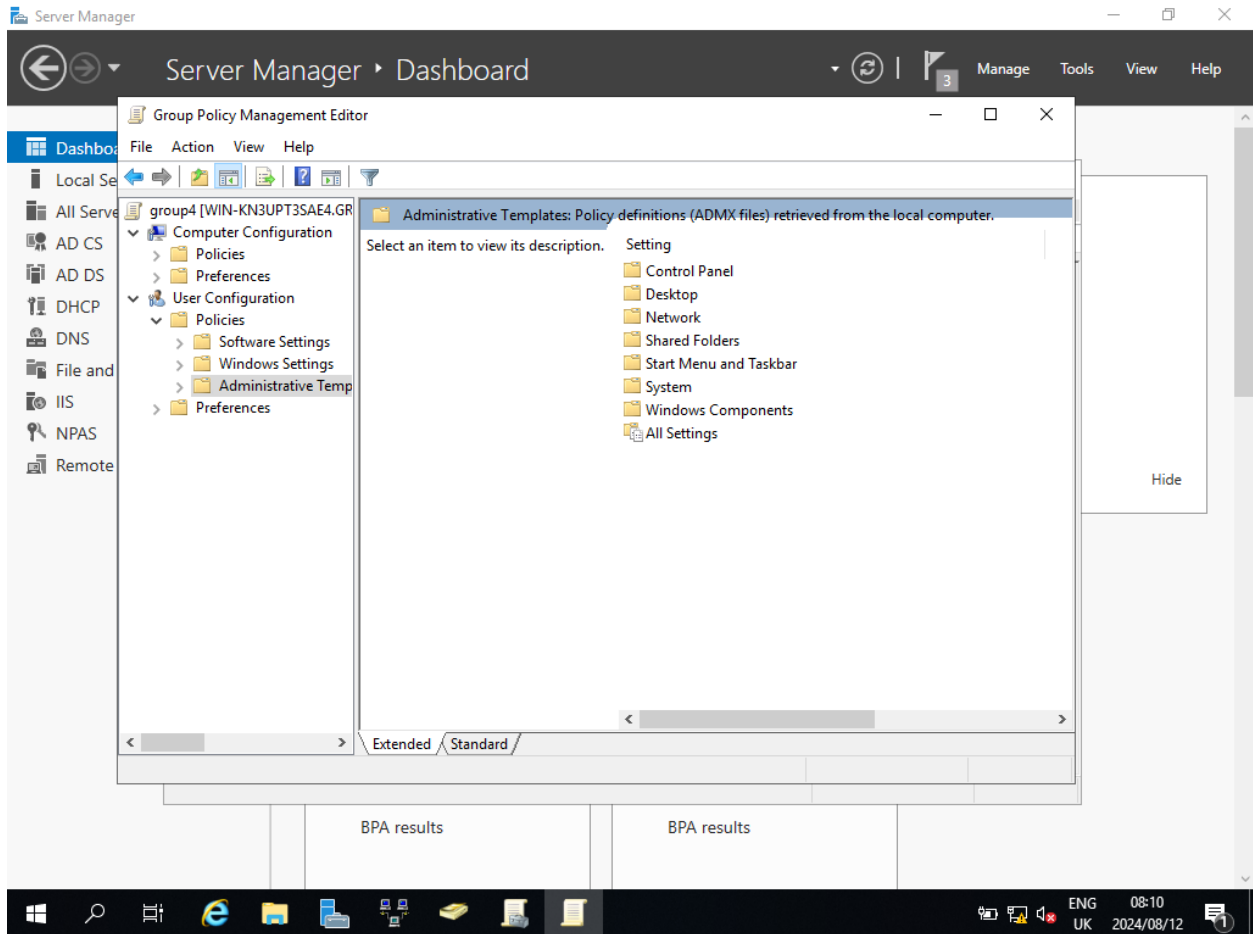
Apply

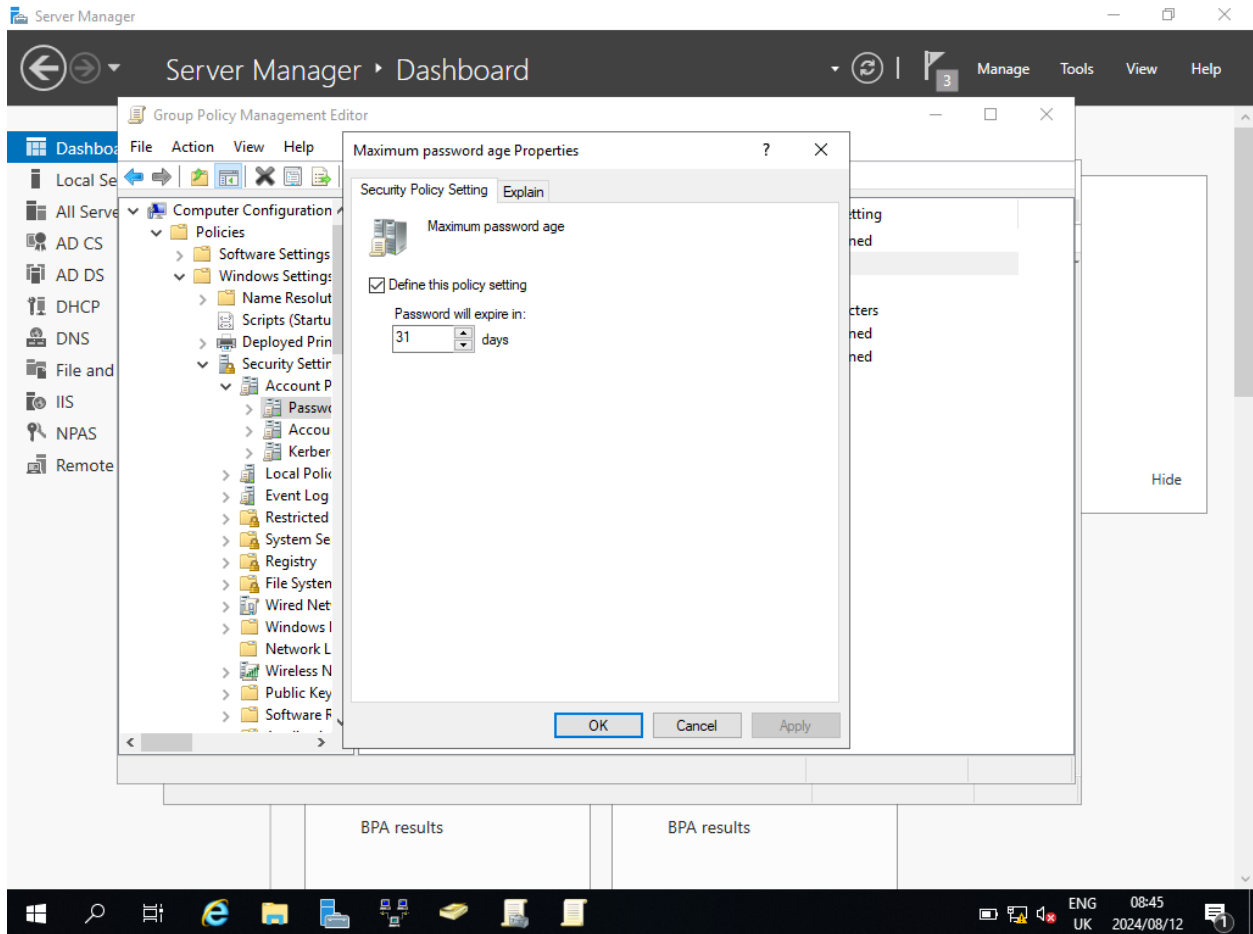
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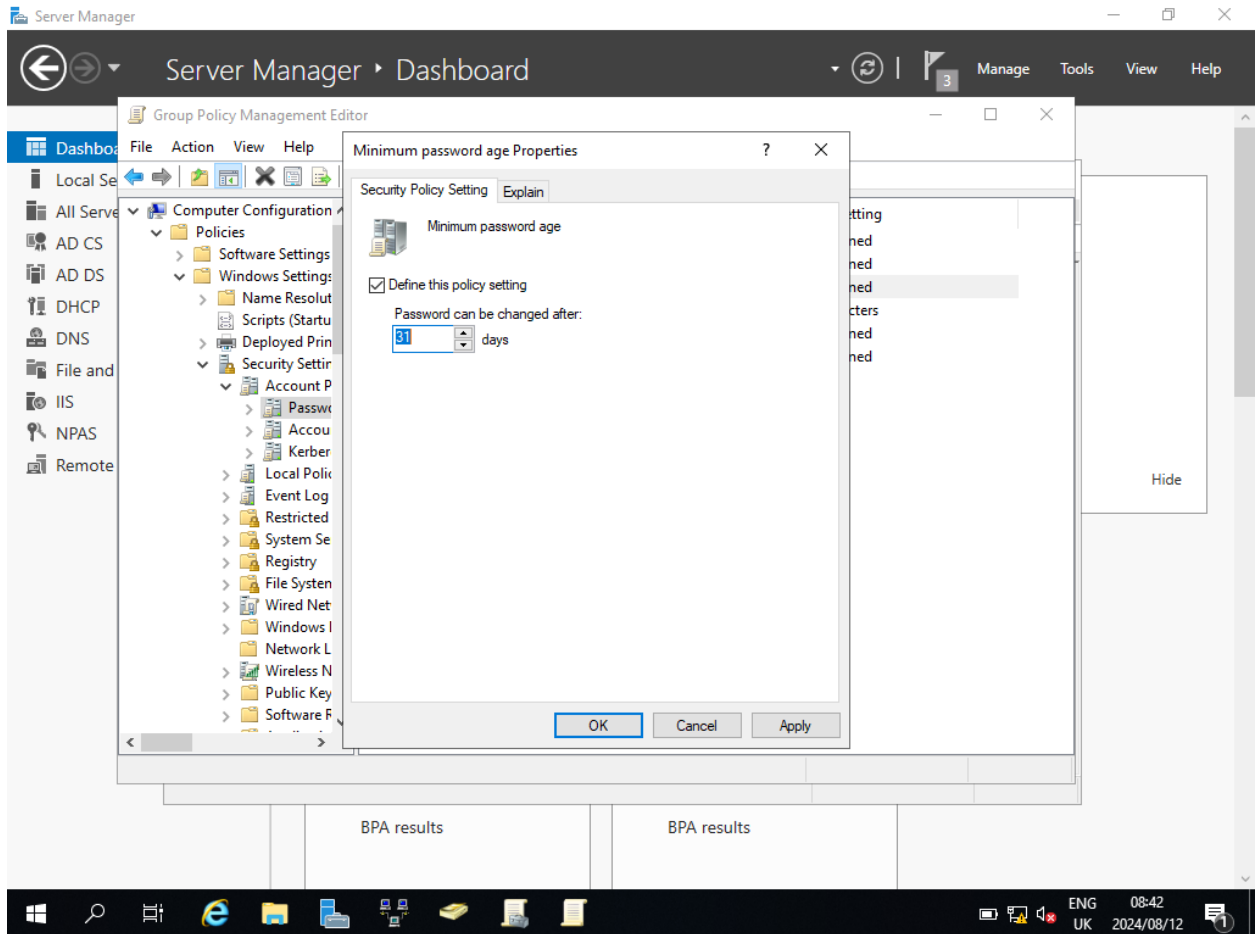
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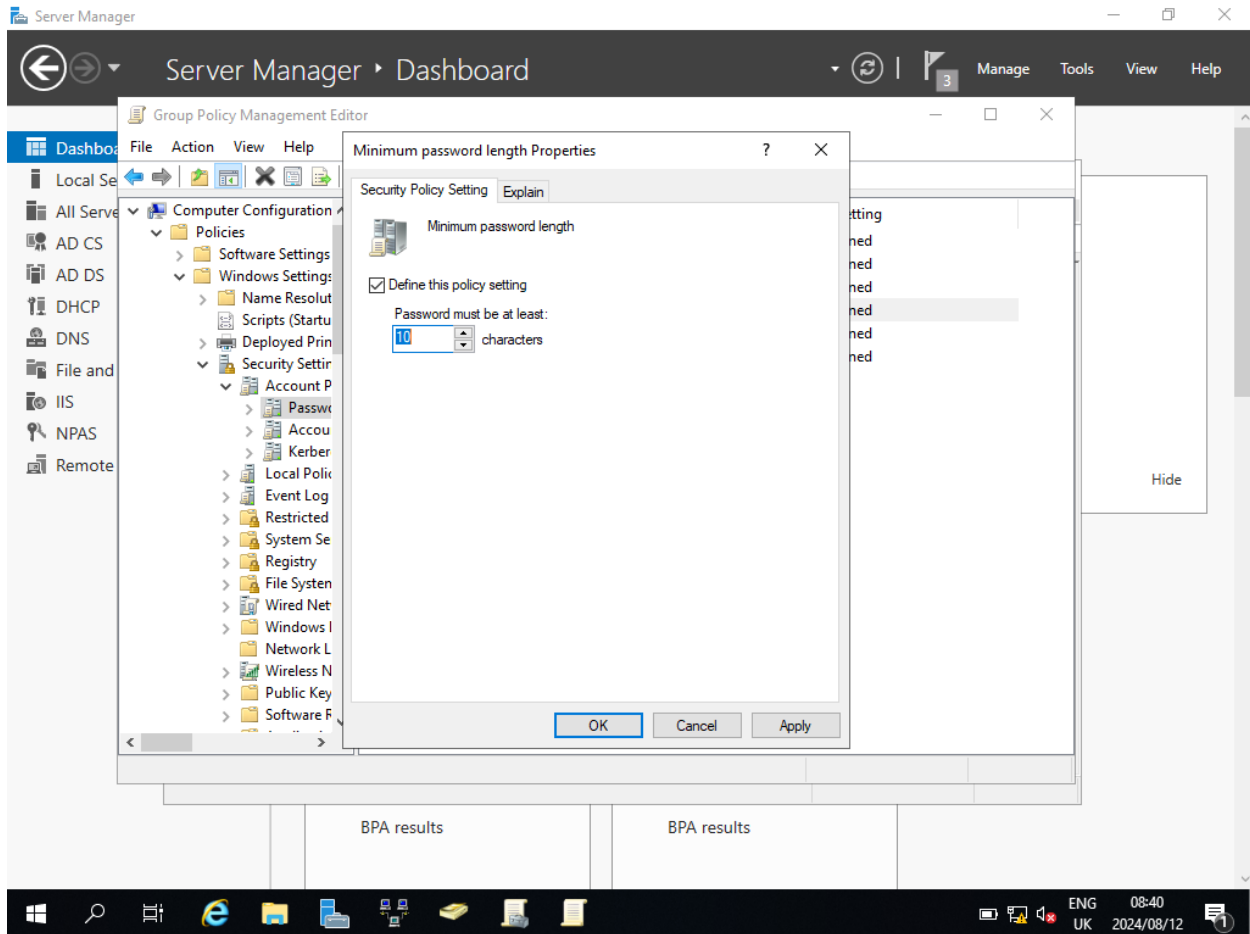
2024/08/12

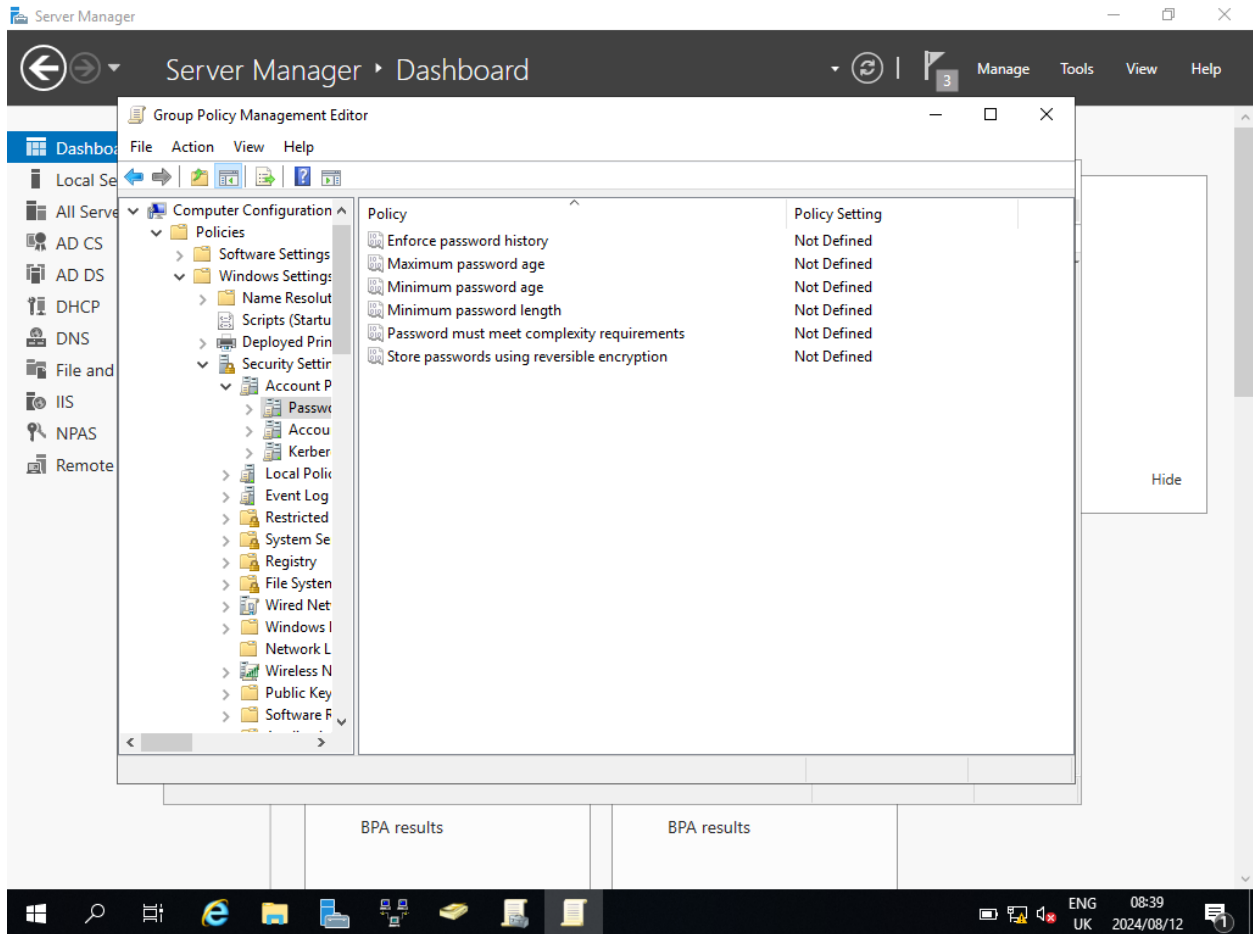


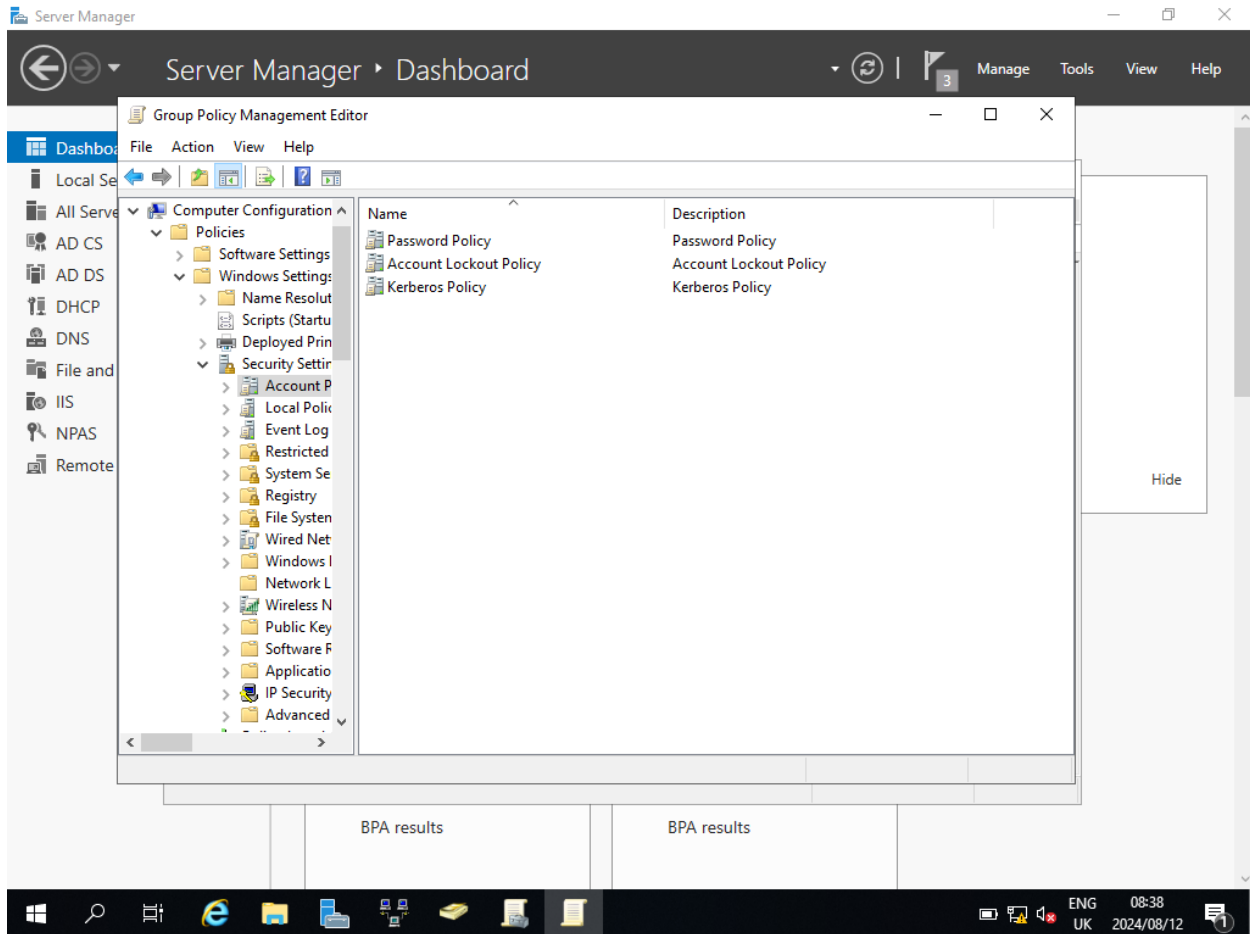












Server Manager

Server Manager Dashboard

Group Policy Management Editor

File Action View Help

Local Server

group4 [WIN-KN3UPT3SAE4]

- Computer Configuration
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 - Deployed Printers
 - Security Settings
 - Policy-based OS Settings
 - Administrative Templates
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 - Printers
 - Server
 - Start Menu and Taskbar
 - System
 - Access-De
 - App-V
 - Audit Proc
 - Credential
 - Device Gu
 - Device He
 - Device Ins

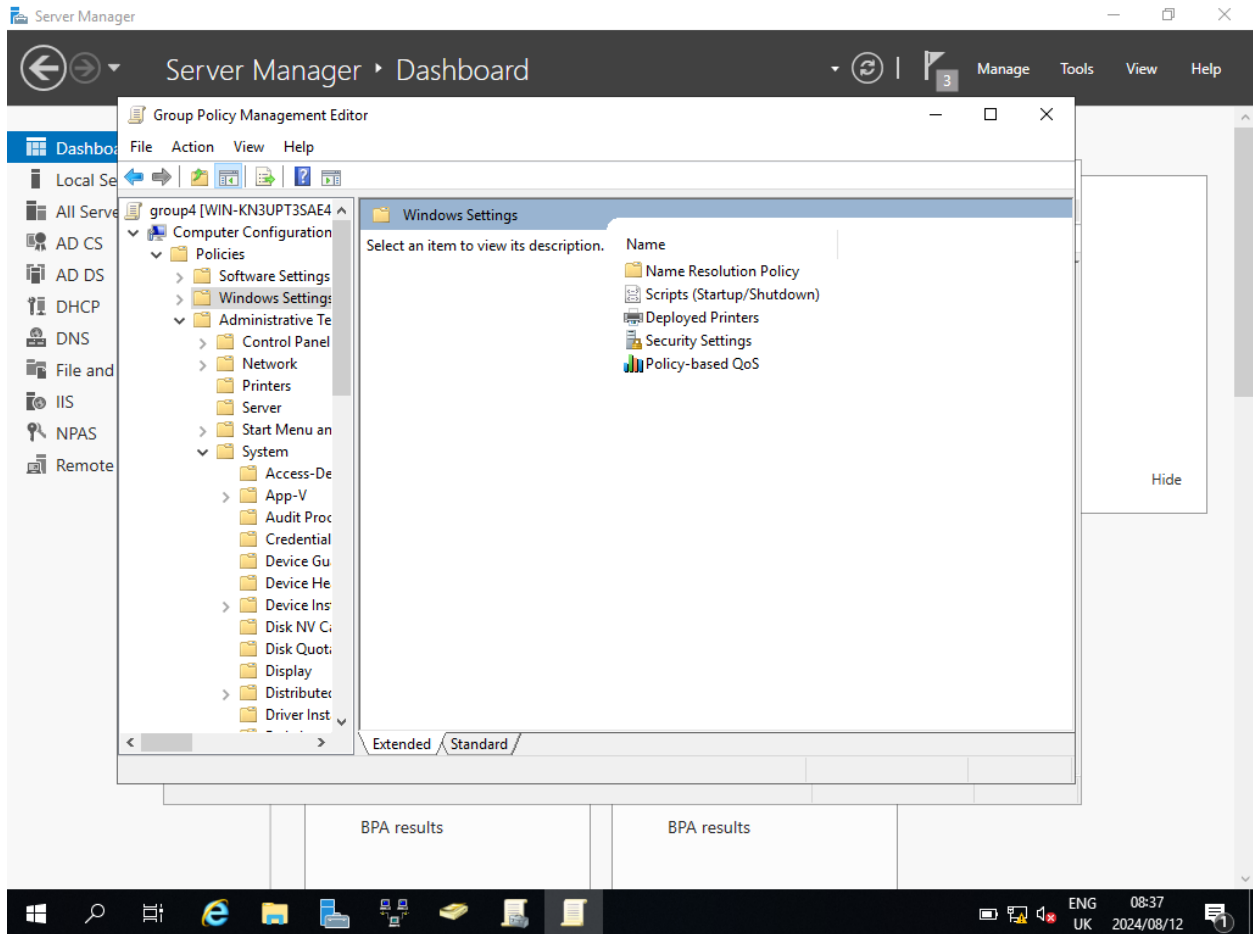
Name	Description
Account Policies	Password and account lockout policies
Local Policies	Auditing, user rights and security options polici...
Event Log	Event Log
Restricted Groups	Restricted Groups
System Services	System service settings
Registry	Registry security settings
File System	File system security settings
Wired Network (IEEE 802.3) Policies	Wired Network Policy Administration. Manage ...
Windows Defender Firewall with Advanc...	Windows Defender Firewall with Advanced Sec...
Network List Manager Policies	Network name, icon and location group policies.
Wireless Network (IEEE 802.11) Policies	Wireless Network Policy Administration. Mana...
Public Key Policies	
Software Restriction Policies	
Application Control Policies	Application Control Policies
IP Security Policies on Active Directory (G...	Internet Protocol Security (IPsec) Administratio...
Advanced Audit Policy Configuration	Advanced Audit Policy Configuration

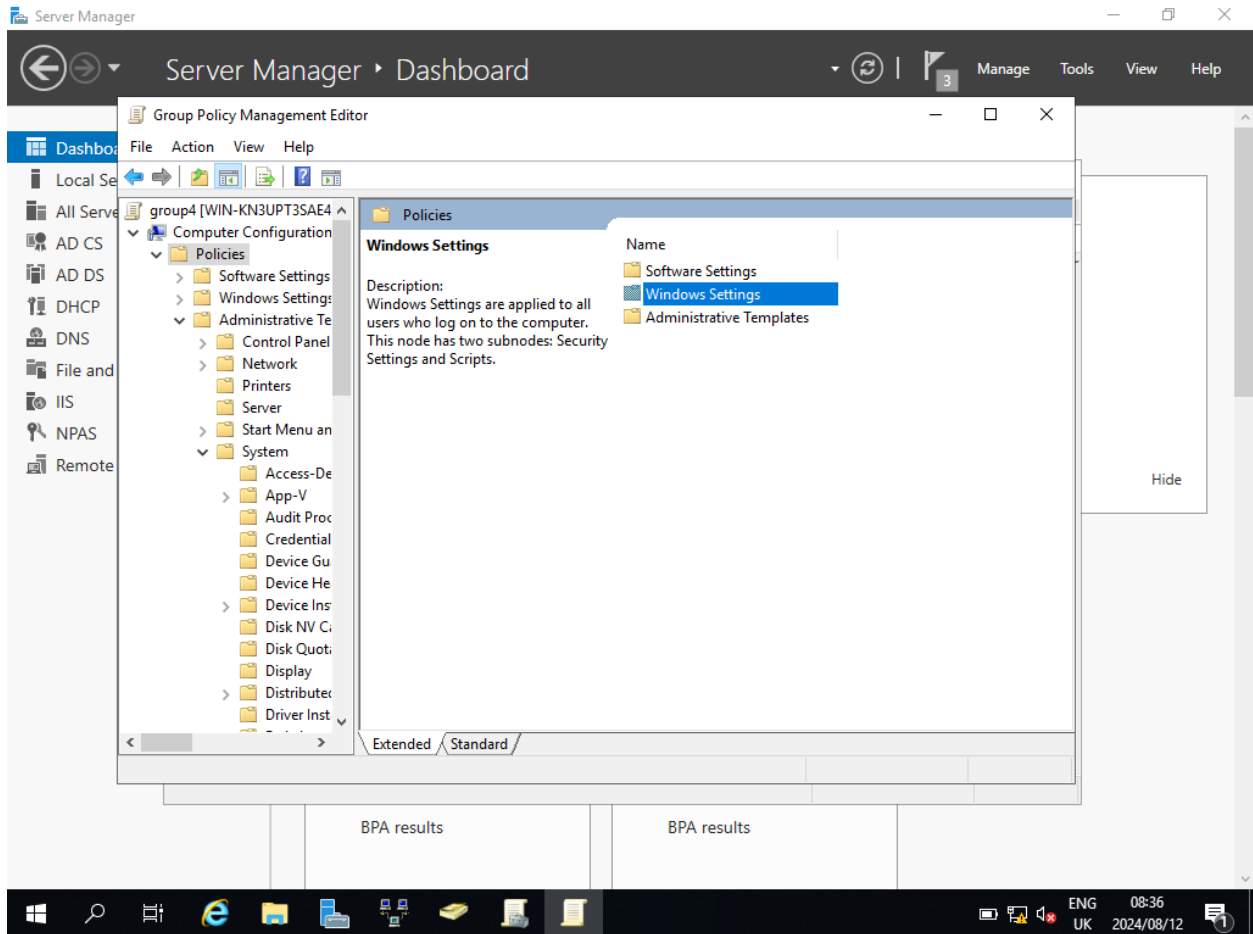
BPA results

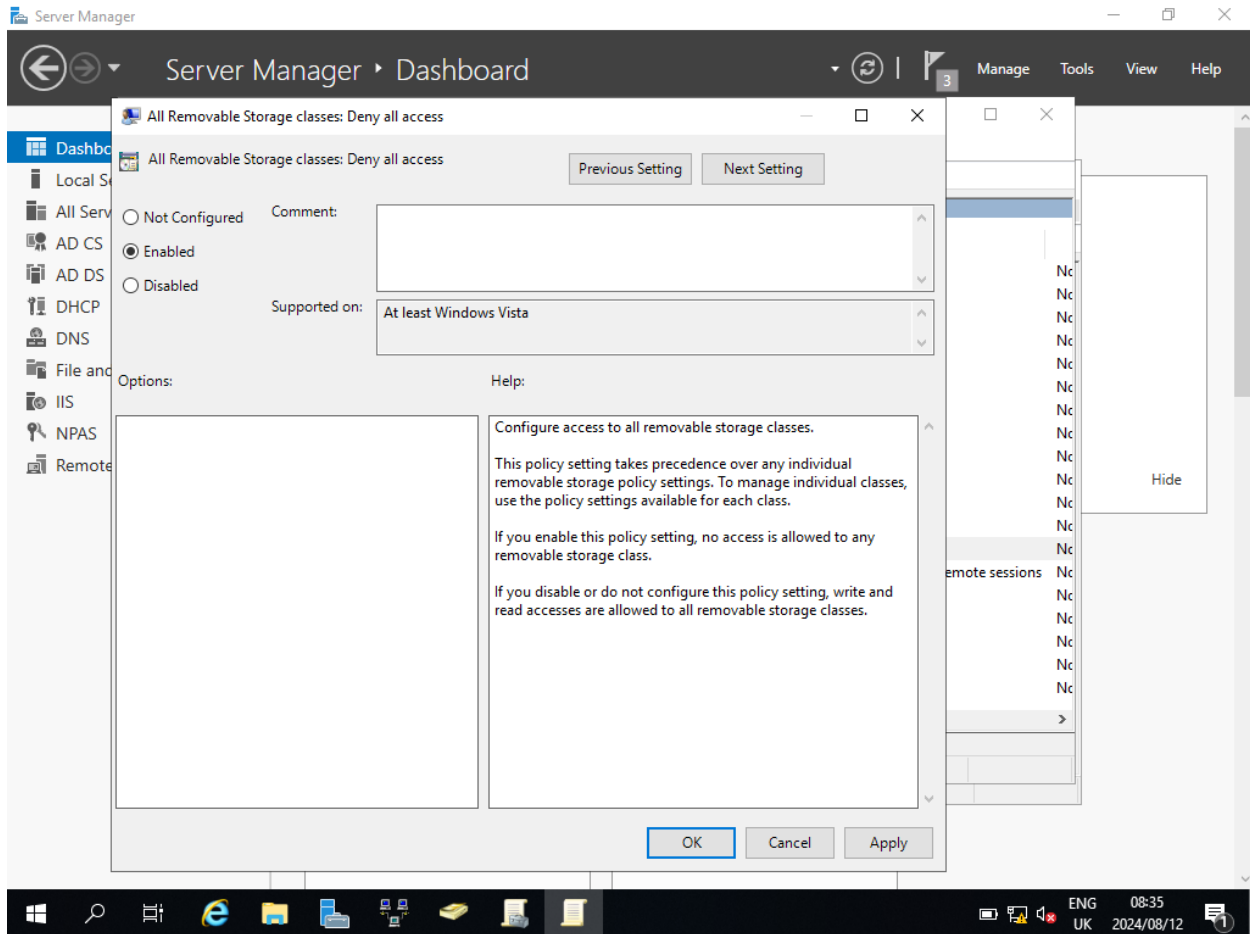
BPA results

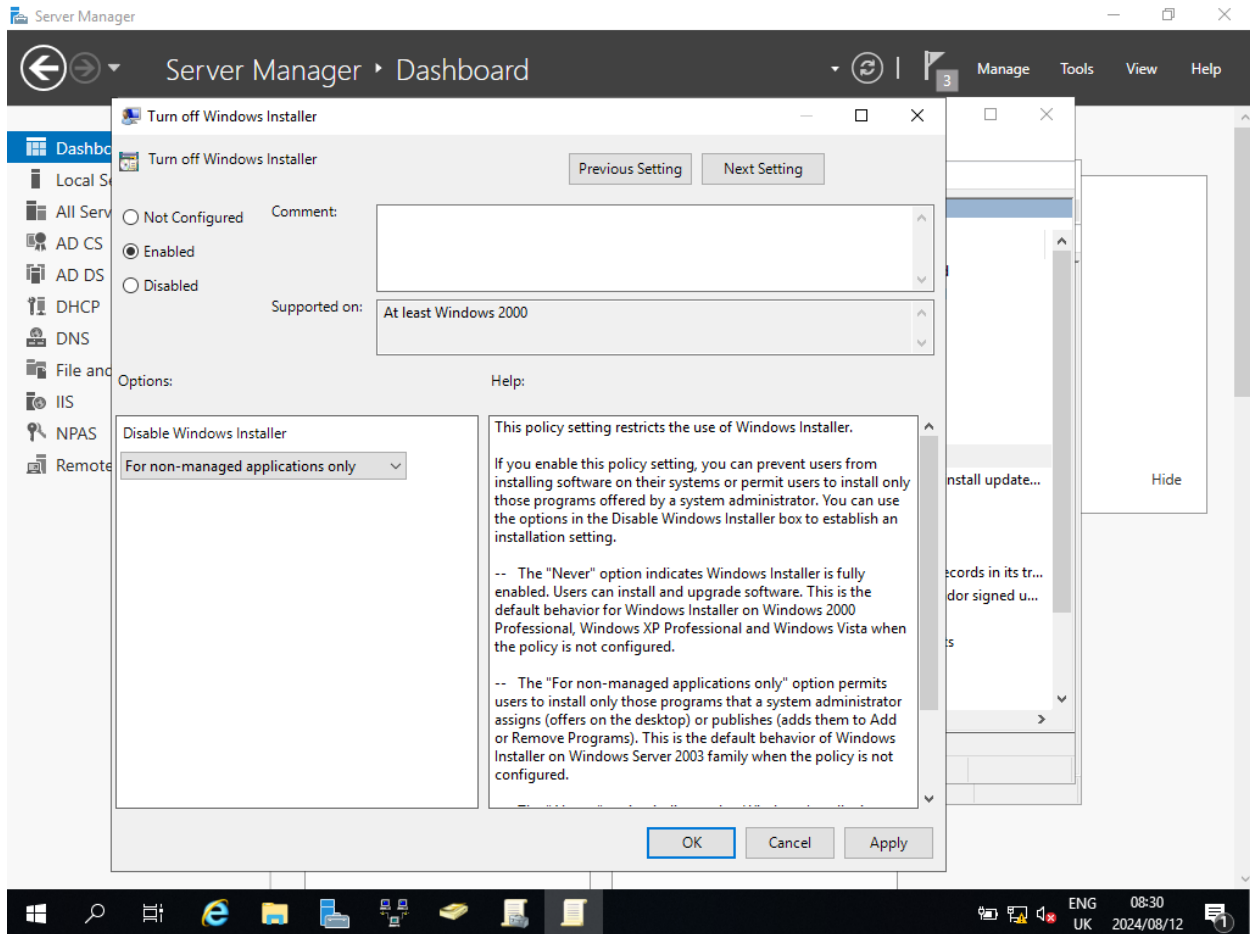
Hide

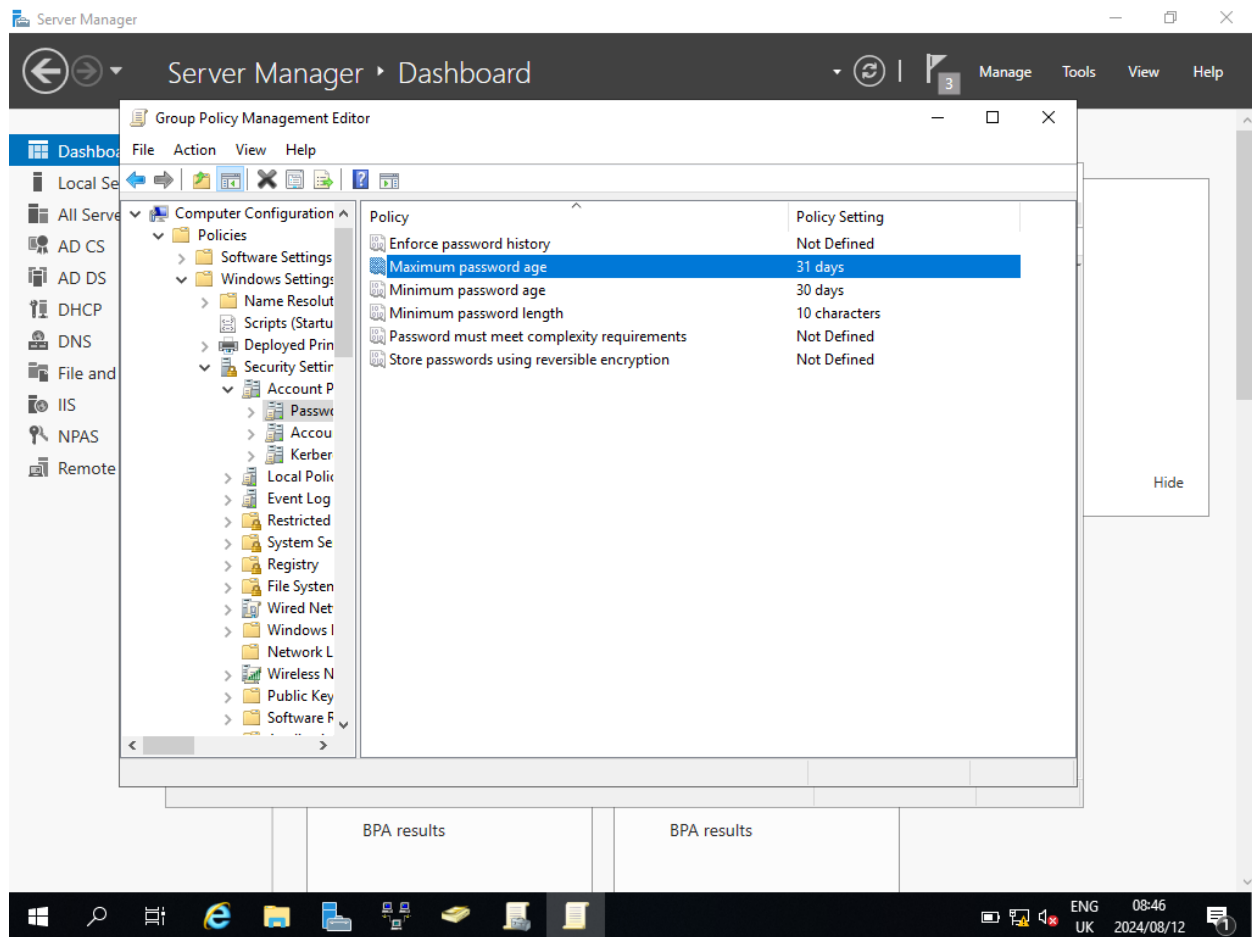
Windows taskbar: ENG 08:37 2024/08/12





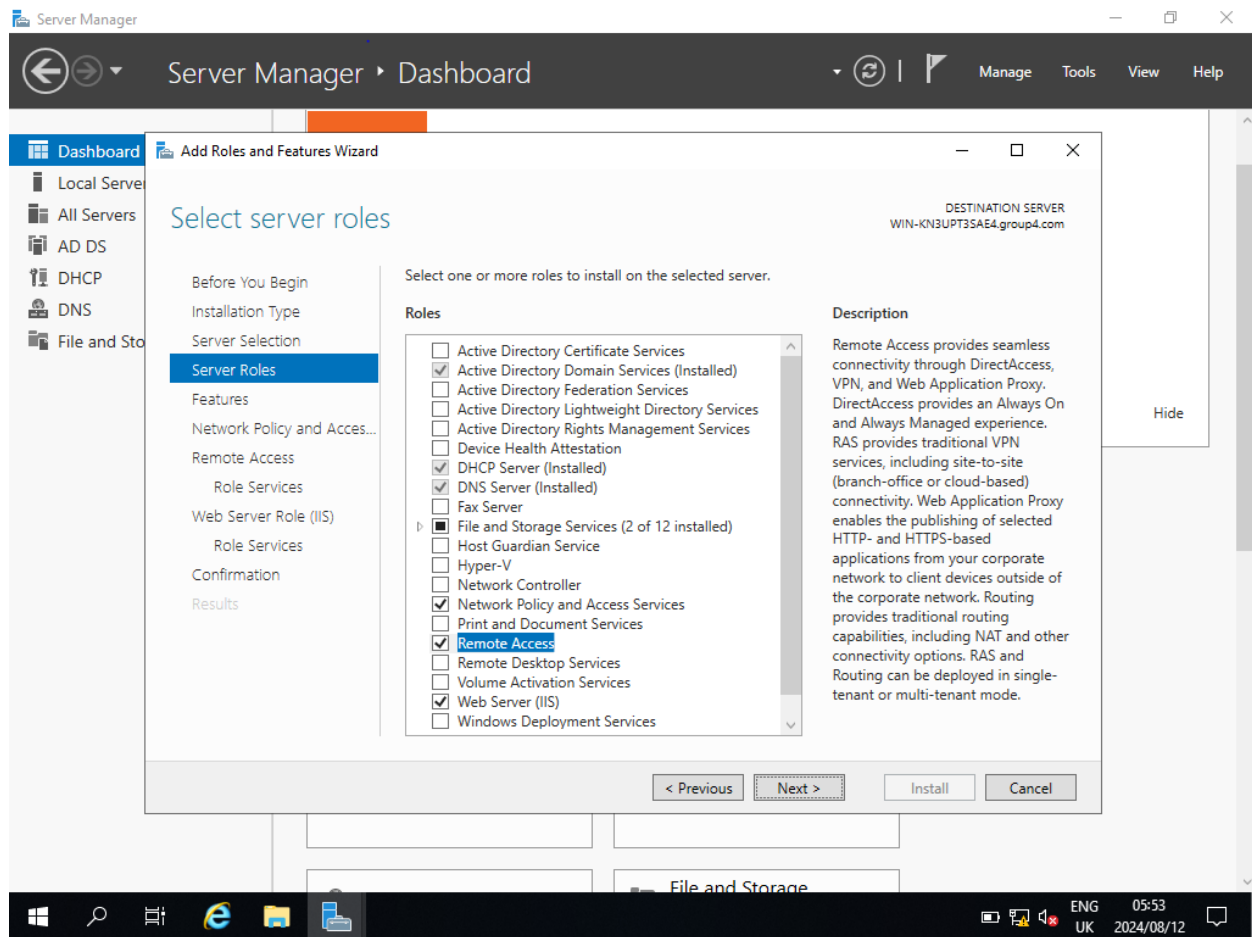






6.Remote access configuration

Remote access allows users to remotely connect to computers and networks. The user can access the desktop, applications, and files while in a different location through remote access. Functions of remote access include Virtual Private Networks (enables a remote user to connect to a corporate network over the internet through secure access), Direct Access (enables a remote user to connect to a corporate network automatically, when they have internet access through seamless connectivity), Remote Desktop Protocol (contain virtual desktops hosted on servers allowing multiple users to access their desktops through virtual desktop infrastructure), Remote Assistance (enables IT support staff to remotely access user's computers to assist with troubleshooting issues).



7.Windows Certificates Service configuration

Windows Certificate Services can be classified as Active Directory Certificate Services. It has the role of providing services that can be customized for managing, distributing, and creating public key certificates that can be used in software security systems. The roles of Windows Certificates are Certification Authority (Issues and manages certificates), Certificate Revocation (Allows for certificates that are no longer trusted to be revoked), Online Responder (Helps clients to verify if a certificate is still valid by providing real-time validation) and Certificate Enrollment (Enables users together with devices to request certification).

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WELCOME TO SERVER MANAGER

Add Roles and Features Wizard

Select role services

DESTINATION SERVER
WIN-KN3UPT3SAE4.group4.com

Before You Begin

Installation Type

Server Selection

Server Roles

Features

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Role Services

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Select the role services to install for Active Directory Certificate Services

Role services

☒ Certification Authority

☒ Certificate Enrollment Policy Web Service

☒ Certificate Enrollment Web Service

☒ Certification Authority Web Enrollment

☐ Network Device Enrollment Service

☒ Online Responder

Description

Online Responder makes certificate revocation checking data accessible to clients in complex network environments.

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Install

Cancel

Hide

Windows Taskbar

ENG UK 07:17 2024/08/12

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Select server roles

Before You Begin

Installation Type

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Server Roles

Features

AD CS

Role Services

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WIN-KN3UPT3SAE4.group4.com

Select one or more roles to install on the selected server.

Roles

☒ Active Directory Certificate Services

☒ Active Directory Domain Services (Installed)

☐ Active Directory Federation Services

☐ Active Directory Lightweight Directory Services

☐ Active Directory Rights Management Services

☐ Device Health Attestation

☒ DHCP Server (Installed)

☒ DNS Server (Installed)

☐ Fax Server

☒ File and Storage Services (2 of 12 installed)

☐ Host Guardian Service

☐ Hyper-V

☐ Network Controller

☒ Network Policy and Access Services (Installed)

☐ Print and Document Services

☒ Remote Access (2 of 3 installed)

☐ Remote Desktop Services

☐ Volume Activation Services

☒ Web Server (IIS) (10 of 43 installed)

☐ Windows Deployment Services

Description

Active Directory Certificate Services (AD CS) is used to create certification authorities and related role services that allow you to issue and manage certificates used in a variety of applications.

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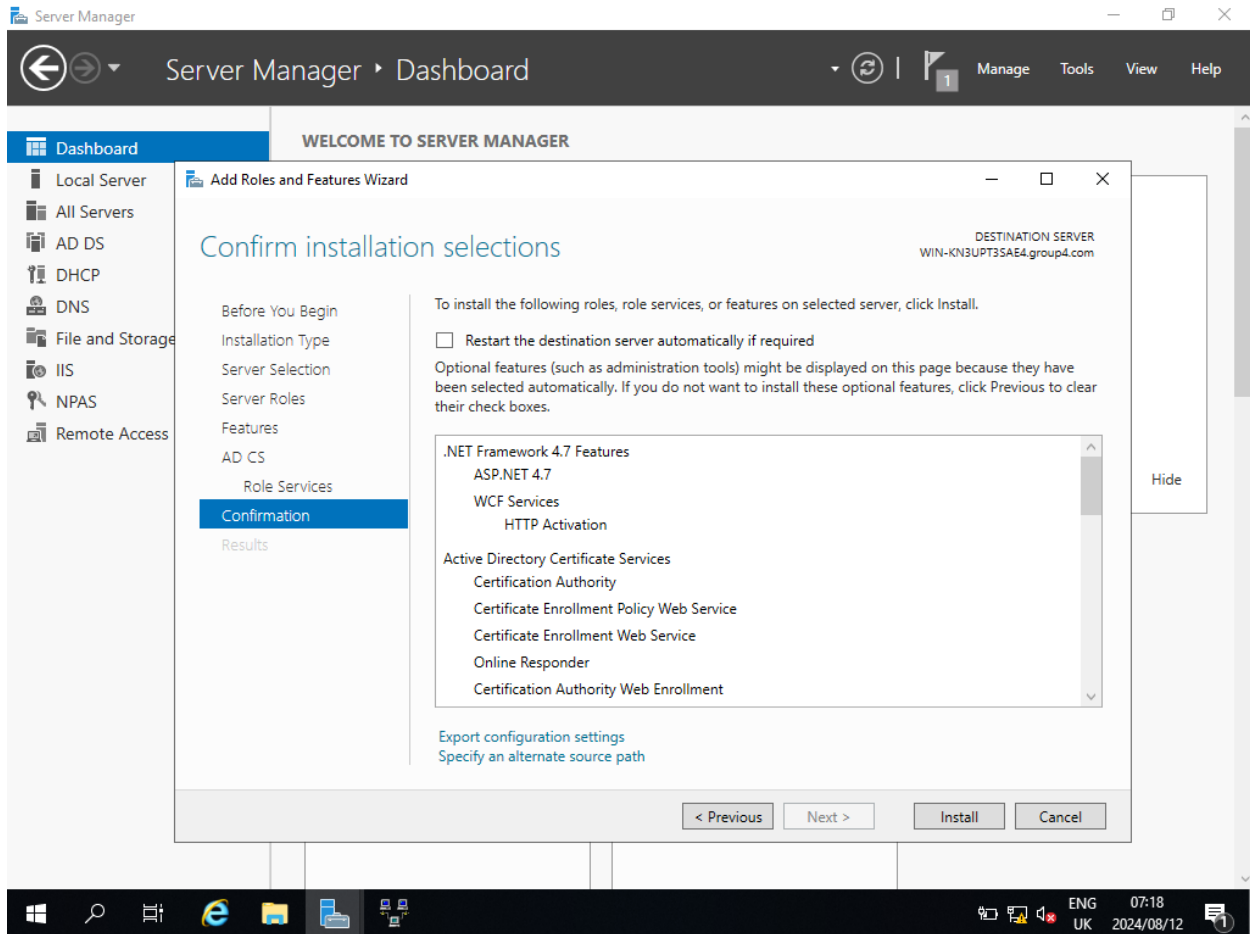
Install

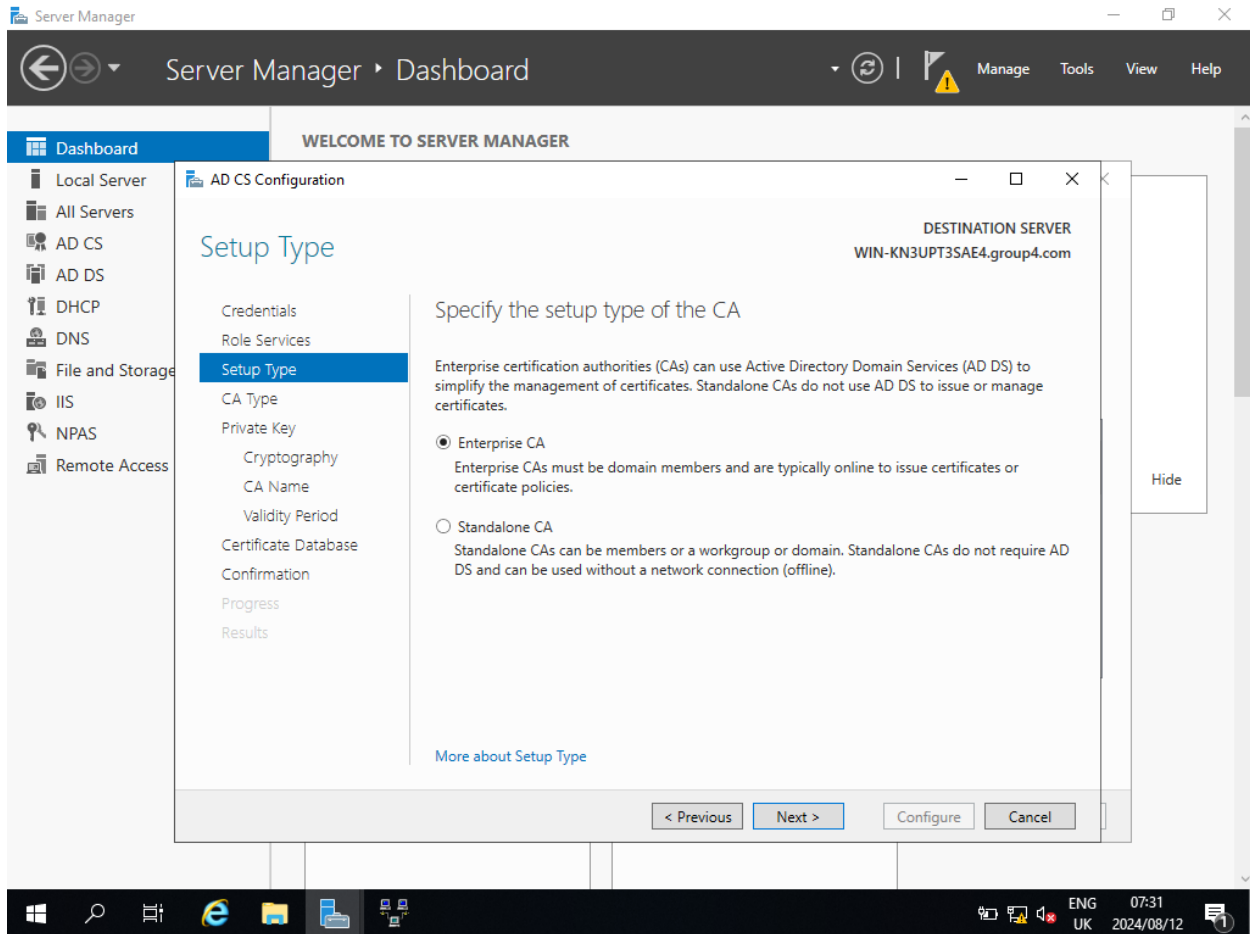
Cancel

Hide

Windows Taskbar

ENG UK 07:14 2024/08/12







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AD CS Configuration

Role Services

Credentials

Role Services

Setup Type

CA Type

Private Key

Cryptography

CA Name

Validity Period

Certificate Database

Confirmation

Progress

Results

Select Role Services to configure

- ☒ Certification Authority
- ☒ Certification Authority Web Enrollment
- ☒ Online Responder
- ☐ Network Device Enrollment Service
- ☐ Certificate Enrollment Web Service
- ☐ Certificate Enrollment Policy Web Service

[More about AD CS Server Roles](#)

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AD CS Configuration

Credentials

Credentials

Role Services

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Results

Specify credentials to configure role services

To install the following role services you must belong to the local Administrators group:

- Standalone certification authority
- Certification Authority Web Enrollment
- Online Responder

To install the following role services you must belong to the Enterprise Administrators group:

- Enterprise certification authority
- Certificate Enrollment Policy Web Service
- Certificate Enrollment Web Service
- Network Device Enrollment Service

Credentials:

Change

[More about AD CS Server Roles](#)

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Add Roles and Features Wizard

Installation progress

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View installation progress



Feature installation

Installation started on WIN-KN3UPT3SAE4.group4.com

.NET Framework 4.7 Features

ASP.NET 4.7

WCF Services

HTTP Activation

Active Directory Certificate Services

Certification Authority

Certificate Enrollment Policy Web Service

Certificate Enrollment Web Service

Online Responder

Certification Authority Web Enrollment

Remote Server Administration Tools

You can close this wizard without interrupting running tasks. View page again by clicking Notifications in the command bar, and

[Export configuration settings](#)

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AD CS Configuration

Private Key

- Credentials
- Role Services
- Setup Type
- CA Type
- Private Key**
- Cryptography
- CA Name
- Validity Period
- Certificate Database
- Confirmation
- Progress
- Results

Specify the type of the private key

To generate and issue certificates to clients, a certification authority (CA) must have a private key.

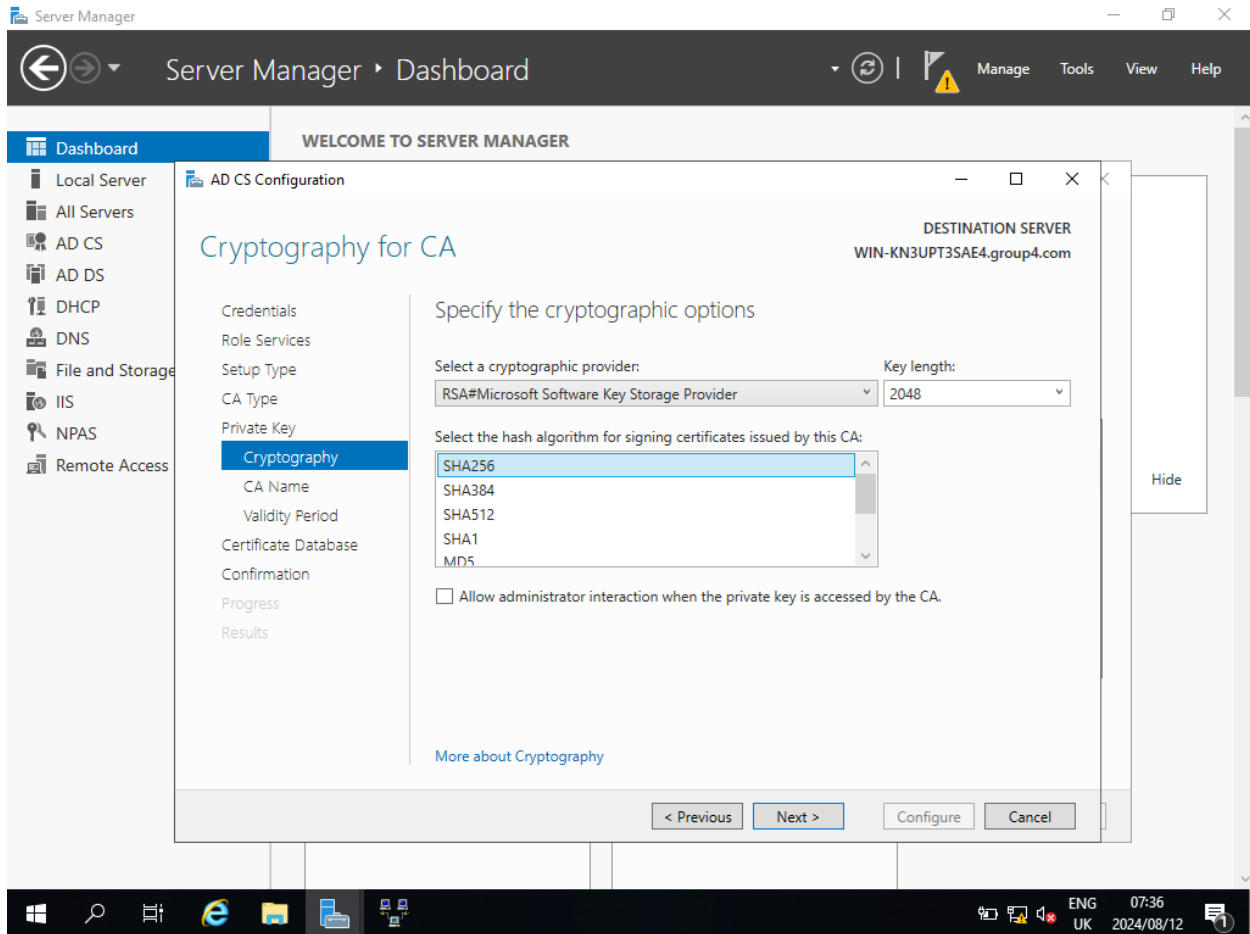
- ☒ Create a new private key
Use this option if you do not have a private key or want to create a new one.
- ☐ Use existing private key
Use this option to ensure continuity with previously issued certificates.
 - ☐ Select a certificate and use its associated private key
Select this option if you have an existing certificate on this computer. Select the certificate to import a certificate and use its associated private key.
 - ☐ Select an existing private key on this computer
Select this option if you have retained private keys from a previous installation. You can use a private key from an alternate source.

[More about Private Key](#)

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CA Type

- Credentials
- Role Services
- Setup Type
- CA Type**
- Private Key
- Cryptography
- CA Name
- Validity Period
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Specify the type of the CA

When you install Active Directory Certificate Services (AD CS), you are creating a public key infrastructure (PKI) hierarchy. A root CA is at the top of the hierarchy and has its own self-signed certificate. A subordinate CA receives a certificate from the root CA.

☒ Root CA

Root CAs are the first and may be the only CAs configured in a PKI hierarchy.

☐ Subordinate CA

Subordinate CAs require an established PKI hierarchy and are authorized to issue certificates for the CA above them in the hierarchy.

[More about CA Type](#)

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Validity Period

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WIN-KN3UPT3SAE4.group4.com

Credentials

Role Services

Setup Type

CA Type

Private Key

Cryptography

CA Name

Validity Period

Certificate Database

Confirmation

Progress

Results

Specify the validity period

Select the validity period for the certificate generated for this certification authority (CA):

5Years

CA expiration Date: 2029/08/12 07:27:00

The validity period configured for this CA certificate should exceed the validity period for the certificates it will issue.

More about Validity Period

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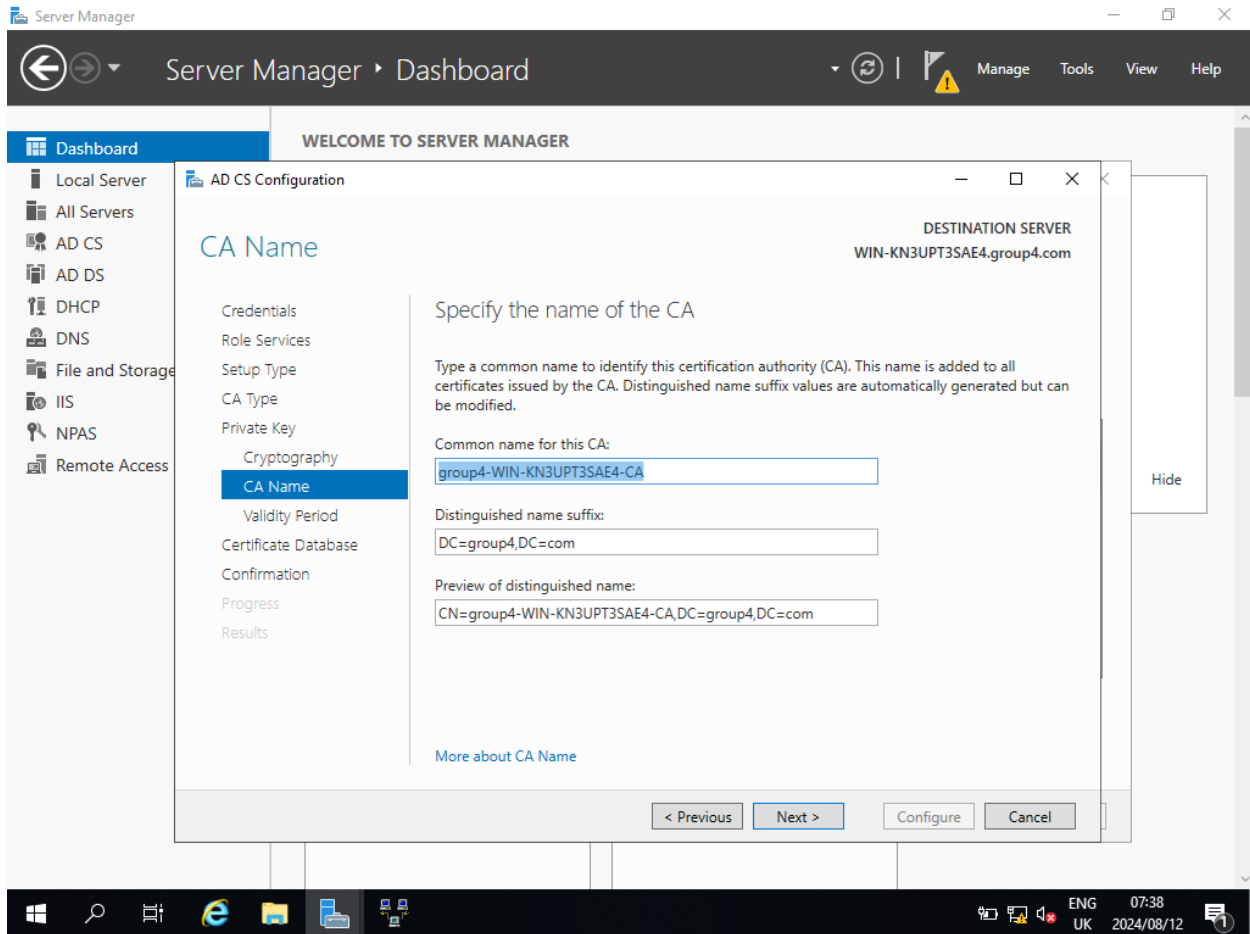
Configure

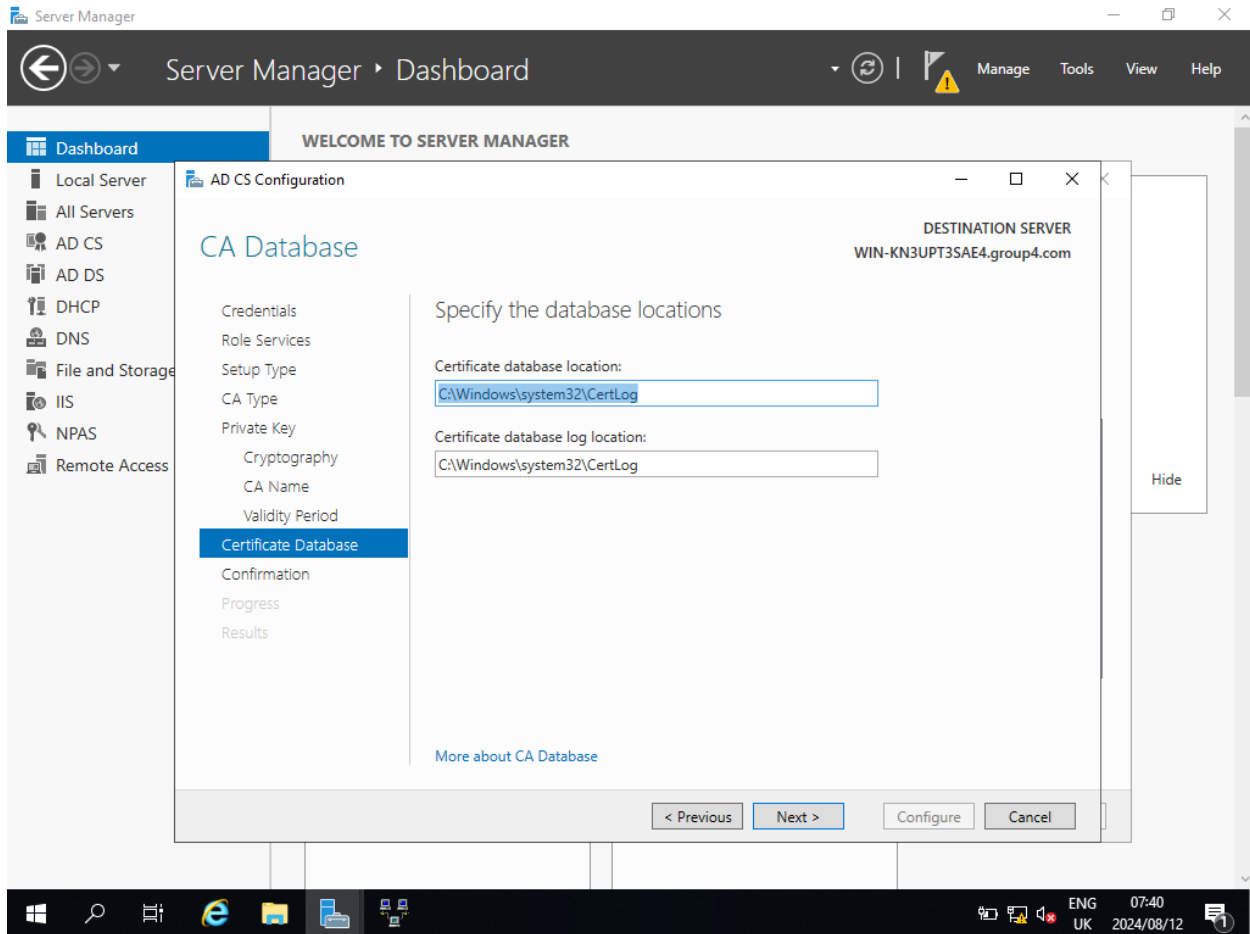
Cancel

Hide

Windows Taskbar

07:39 2024/08/12





Server Manager

Server Manager ▸ Dashboard

ManageToolsViewHelp

Back to DNSDashboard

Local ServerAll ServersAD CSAD DSDHCPDNSFile and StorageIISNPASRemote Access

AD CS Configuration

Results

WELCOME TO SERVER MANAGER

DESTINATION SERVER
WIN-KN3UPT3SAE4.group4.com

Results

Credentials

Role Services

Setup Type

CA Type

Private Key

Cryptography

CA Name

Validity Period

Certificate Database

Confirmation

Progress

Results

The following roles, role services, or features were configured:

Active Directory Certificate Services

Certification Authority

More about CA Configuration

Configuration succeeded

Certification Authority Web Enrollment

More about Web Enrollment Configuration

Configuration succeeded

Online Responder

More about OCSP Configuration

Configuration succeeded

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Close

Cancel

Hide

Windows Taskbar

ENG UK07:452024/08/12



Dashboard

WELCOME TO SERVER MANAGER

- Local Server
- All Servers
- AD CS
- AD DS
- DHCP
- DNS
- File and Storage
- IIS
- NPAS
- Remote Access

AD CS Configuration

Confirmation

- Credentials
- Role Services
- Setup Type
- CA Type
- Private Key
 - Cryptography
- CA Name
- Validity Period
- Certificate Database
- Confirmation**
- Progress
- Results

To configure the following roles, role services, or features, click Config

Active Directory Certificate Services

Certification Authority

CA Type:	Enterprise Root
Cryptographic provider:	RSA#Microsoft Software Key Storage
Hash Algorithm:	SHA256
Key Length:	2048
Allow Administrator Interaction:	Disabled
Certificate Validity Period:	2029/08/12 07:27:00
Distinguished Name:	CN=group4-WIN-KN3UPT3SAE4-CA
Certificate Database Location:	C:\Windows\system32\CertLog
Certificate Database Log Location:	C:\Windows\system32\CertLog

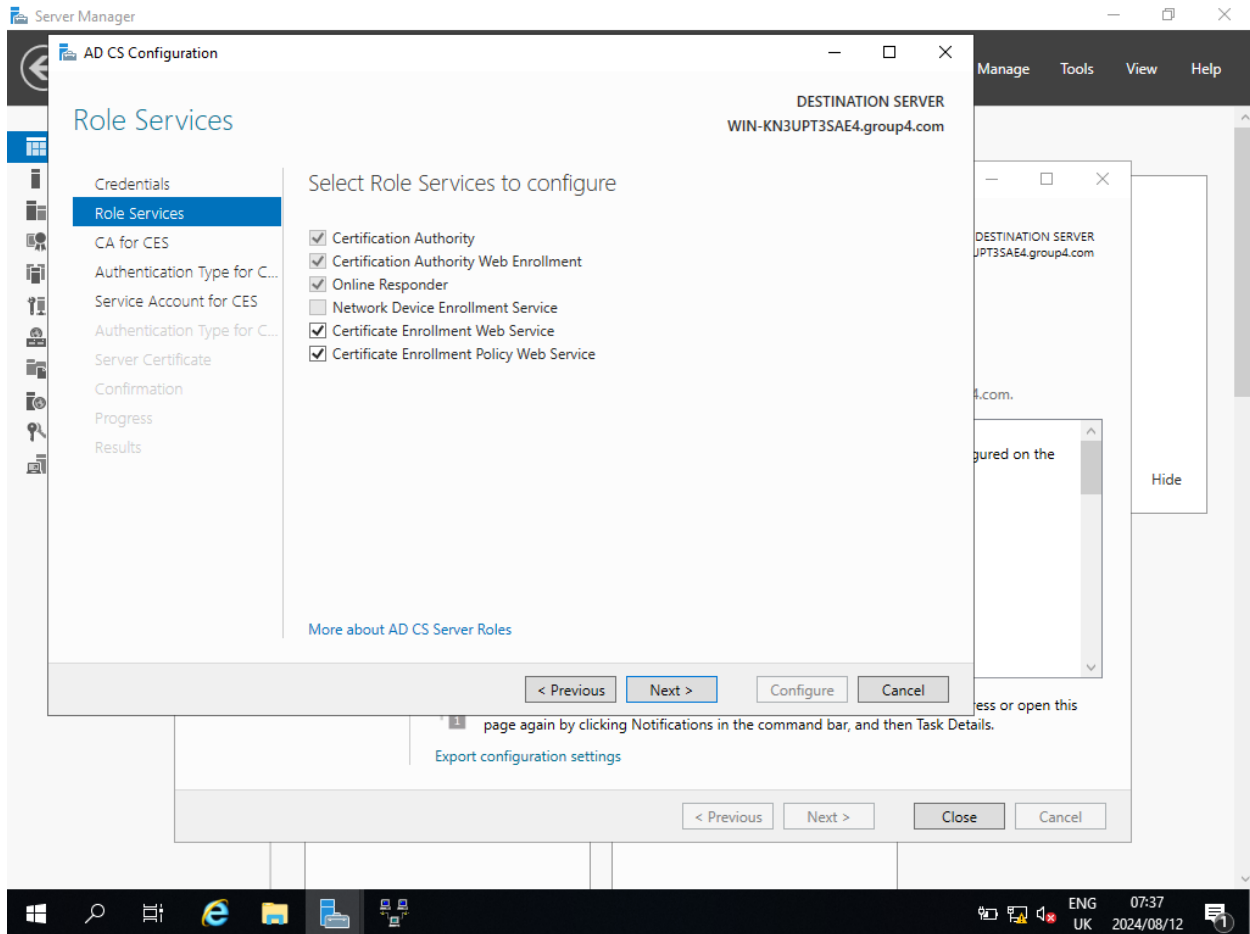
Certification Authority Web Enrollment

Online Responder

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Credentials

DESTINATION
WIN-KN3UPT3SAE4.gro

Credentials

Role Services

Confirmation

Progress

Results

Specify credentials to configure role services

To install the following role services you must belong to the local Administrators group:

- Standalone certification authority
- Certification Authority Web Enrollment
- Online Responder

To install the following role services you must belong to the Enterprise Admins group:

- Enterprise certification authority
- Certificate Enrollment Policy Web Service
- Certificate Enrollment Web Service
- Network Device Enrollment Service

Credentials: [More about AD CS Server Roles](#)

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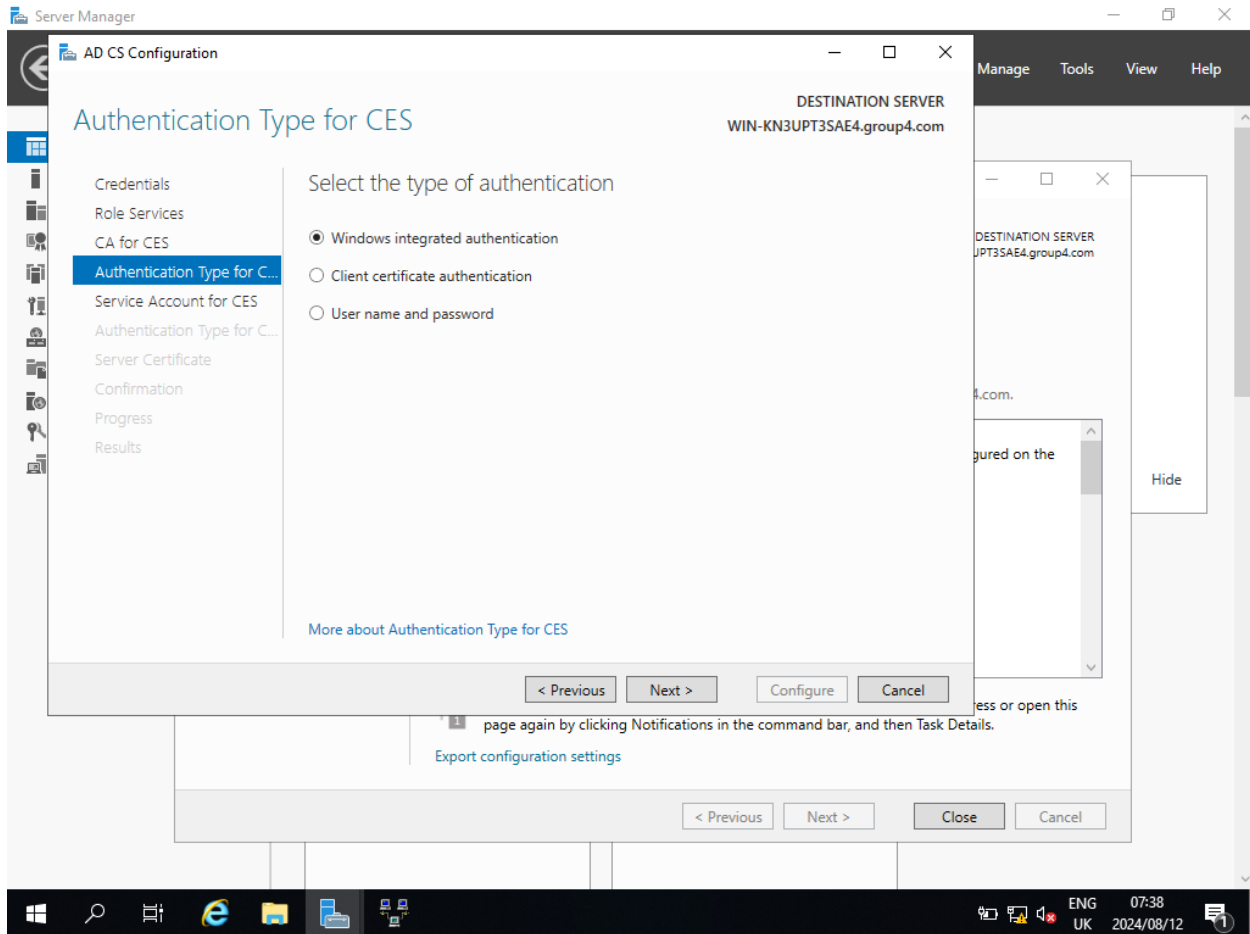
Configure

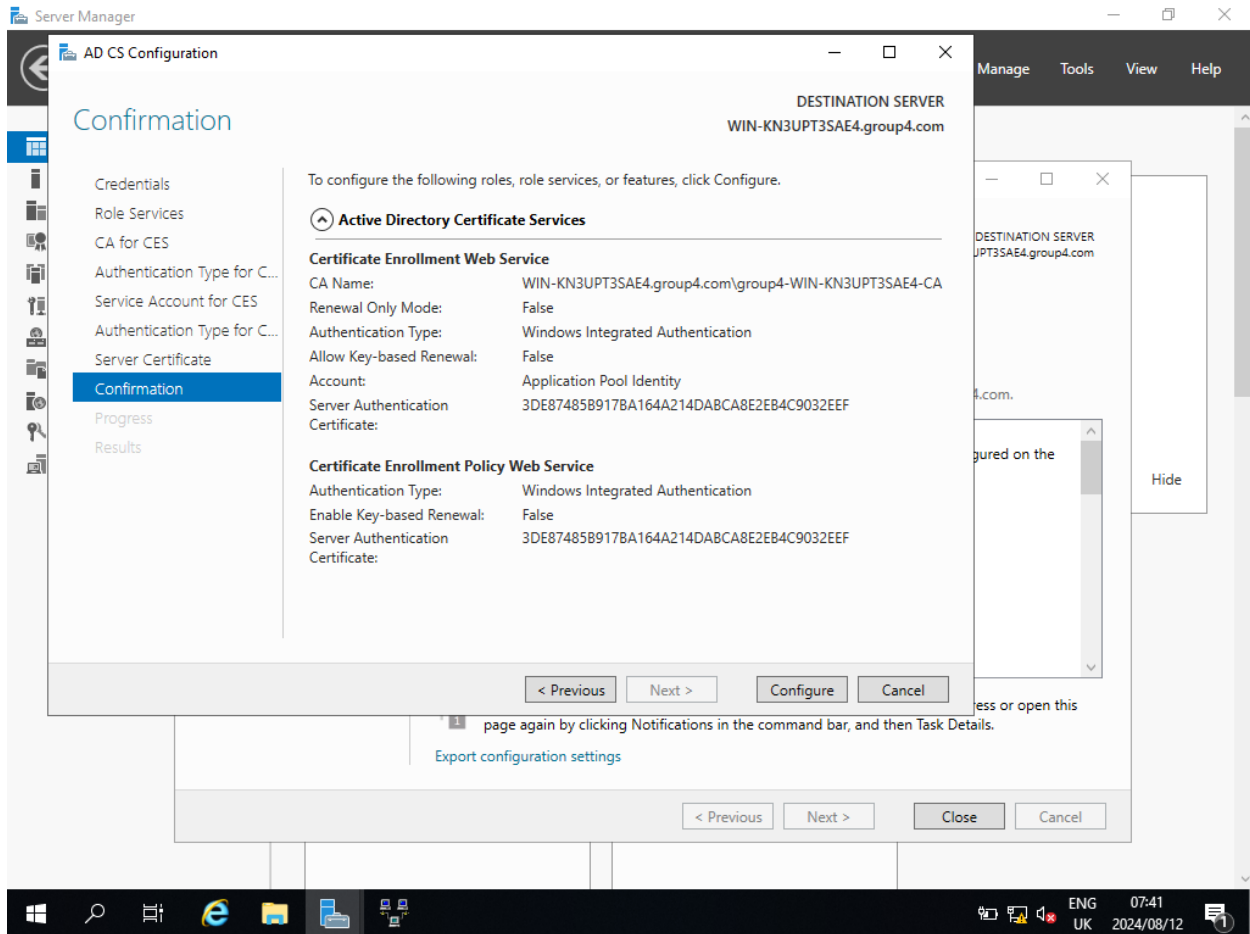
1 page again by clicking Notifications in the command bar, and
[Export configuration settings](#)

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Server Certificate

DESTINATION
WIN-KN3UPT3SAE4.gro

- Credentials
- Role Services
- CA for CES
- Authentication Type for C...
- Service Account for CES
- Authentication Type for C...
- Server Certificate**
- Confirmation
- Progress
- Results

Specify a Server Authentication Certificate

When communicating with clients, the web service(s) uses Secure Sockets Layer (SSL) protocol to encrypt network traffic.

- ☒ Choose an existing certificate for SSL encryption (recommended)

Issued To	Issued By	Expiration Date
group4-WIN-KN3UPT3SAE4-CA	group4-WIN-KN3UPT3SAE4-CA	2029/08/12

Properties

Refresh

- ☐ Choose and assign a certificate for SSL later

 For this role service to function, you must configure this server with a valid certificate.

[More about Server Certificate](#)

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Configure

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Results

DESTINATION
WIN-KN3UPT3SAE4.gro

Credentials

Role Services

CA for CES

Authentication Type for C...

Service Account for CES

Authentication Type for C...

Server Certificate

Confirmation

Progress

Results

The following roles, role services, or features were configured:

Active Directory Certificate Services

Certificate Enrollment Web Service

✓ Configuration succeeded

i Delegation must be enabled for the web service account when the Certificate Enrollment Service is installed and all of the following conditions apply:

1. The Certificate Enrollment Web Service is installed on a separate computer from the certification authority
2. Renewal-only mode is not enabled
3. The authentication type is set for Kerberos or Certificate Authentication

[More about CES Configuration](#)

Certificate Enrollment Policy Web Service

✓ Configuration succeeded

i Before clients can use this web service, a server authentication certificate must be con to encrypt communication between clients and the service. Use the IIS snap-in to verify server authentication certificate.

i Before clients can use the Certificate Enrollment Policy Web service, Group Policy setti be applied to their computers to direct certificate enrollment requests to the web serv

[More about CEP Configuration](#)

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Progress

- Credentials
- Role Services
- CA for CES
- Authentication Type for C...
- Service Account for CES
- Authentication Type for C...
- Server Certificate
- Confirmation
- Progress**
- Results

The following roles, role services, or features are being configured:

Configuring...

Active Directory Certificate Services

Certificate Enrollment Web Service

Certificate Enrollment Policy Web Service

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Configure

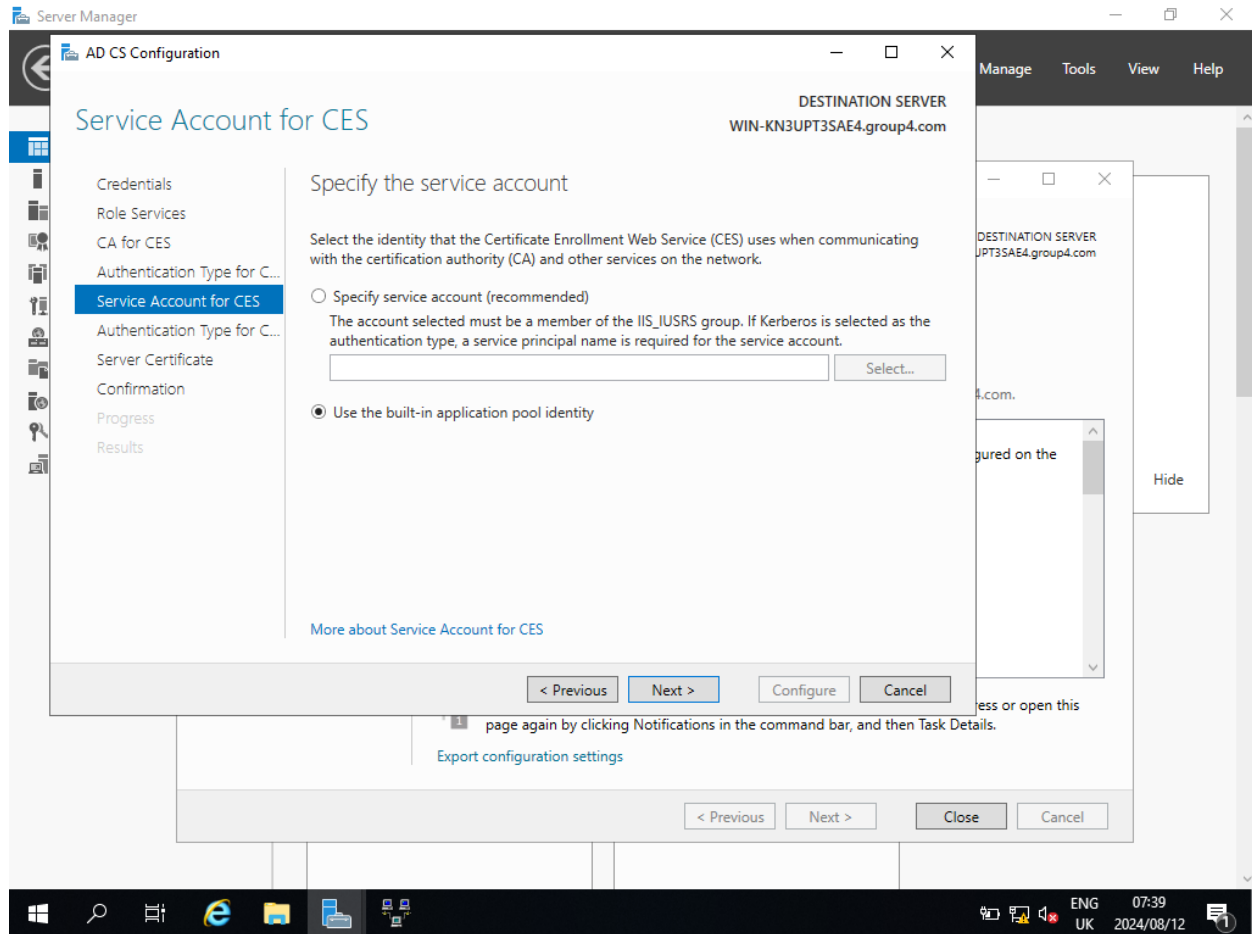
page again by clicking Notifications in the command bar, and

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8.Conclusion

This project set out to design and implement a Windows Server 2019 environment capable of supporting the growing infrastructure needs of an organization, and this objective was achieved through the successful installation and configuration of the core server roles required for a functional enterprise network. The installation of Windows Server 2019 provided the foundation on which Active Directory Domain Services was deployed, enabling centralized management of network resources, users, and security through a structured directory hierarchy. Building on this, DHCP configuration automated the assignment of IP addresses across the network, reducing the administrative overhead and configuration errors associated with manual addressing.

Group Policy configuration strengthened the security posture of the environment by enforcing restrictions on Command Prompt access, removable media usage, and unauthorized software installations, while also implementing stronger password policies through increased minimum password length and reduced maximum password age. These

measures directly address common attack vectors such as privilege escalation, malware introduction via external drives, and credential compromise through weak or long-lived passwords. Remote Access configuration further extended the usability of the environment by allowing authorized users to securely connect to network resources from outside locations, while Windows Certificate Services enabled the issuing and management of digital certificates to support secure authentication and communication across the network.

Overall, the project demonstrated that the individual server roles do not function in isolation but work together as an integrated system: Active Directory provides the identity and access backbone, DHCP and DNS support network connectivity and name resolution, Group Policy enforces organization-wide security standards, and Remote Access and Certificate Services extend secure connectivity beyond the local network. The practical implementation of these services provided valuable hands-on experience in enterprise systems administration and reinforced the theoretical concepts covered in the module, while also highlighting the importance of layered security controls in protecting organizational IT infrastructure as it scales.

9.Reference

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